Attitude Control for Accurate Ground Station Pointing of Bias Momentum Geosynchronous Spacecraft Amid Orbital Deviations

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Abstract: Geosynchronous spacecraft subject to orbital deviations require precise attitude control to maintain alignment with designated ground targets. This study investigates the dynamics and control of bias-momentum geosynchronous spacecraft, emphasizing their suitability for precision pointing applications. Perturbations in the sub-satellite point (SSP) due to deviations from an ideal geosynchronous orbit were simulated, offering critical insights for satellite tracking, communication link stability, and Earth observation operations.

Attitude guidance commands were derived using SSP perturbation data and target coordinates and a roll-actuated control law was employed to execute the guidance commands, with particular attention given to mitigating challenges in pitch measurement accuracy. A mathematical model of the Earth sensor was developed to enhance simulation fidelity, and control signals were converted into thruster activation commands via pulse-width pulse-frequency modulation. The proposed methodology demonstrates robust control strategies for maintaining precision pointing under orbital deviations.

Keywords: Attitude Control, Precision Pointing, Bias-momentum Satellites, Geosynchronous Spacecraft, Orbital Deviations, Guidance Commands, Roll-actuated Control, Sub-satellite Point, Sensor Characterization, Thruster Modeling, Pulse-width Pulse-frequency Modulation.

1 Introduction

Geostationary orbits are essential for communication and Earth observation, providing a stationary position relative to Earth. However, launch inaccuracies and environmental perturbations can cause orbital deviations, leading to variations in the sub-satellite point (SSP) and affecting pointing accuracy and mission performance. Maintaining precise pointing is crucial for ensuring data quality and connectivity in Earth observation, remote sensing, and communication missions. To mitigate SSP perturbations, robust attitude control strategies are necessary, typically involving on-board systems such as reaction wheels, magnetic torquers, or thrusters. Momentum bias satellites, which use spinning momentum wheels, offer an efficient solution by maintaining pointing direction with minimal maneuvers, making them well-suited for long-duration missions.

This study focuses on the development of attitude control strategies for bias-momentum geosynchronous spacecraft, aiming to reduce pointing misalignments caused by orbital deviations. Through analytical formulations and simulations, we demonstrate that precision pointing can be achieved using moderate-sized momentum wheels and offset thrusters. Additionally, sensor models are developed to enhance the accuracy of simulations, enabling effective decision-making for attitude control and ensuring reliable alignment with ground stations under operational constraints.

2 Latitude-Longitude Perturbations of the Sub-Satellite Point

Geostationary orbits, approximately 42,164 km from Earth's center [1], enable satellites to maintain a fixed sub-satellite point (SSP) on the Earth's surface, critical for communication and Earth observation missions. Geosynchronous orbits, which may have non-zero inclination and eccentricity, result in SSP oscillations due to orbital perturbations and deviations from ideal parameters. These perturbations in latitude and longitude necessitate precise guidance commands to maintain target alignment.

Analyzing SSP deviations is essential for optimizing orbital maneuvers, mission planning, and control strategies. This study addresses the impact of orbital deviations on SSP perturbations and presents methodologies for achieving precision pointing in geostationary and geosynchronous satellite missions.

2.1 Inclination-Induced Perturbations

The non-zero inclination of a spacecraft induces variations in the latitude and longitude of its sub-satellite point (SSP). Geocentric latitude and longitude are derived from the satellite's orbital parameters and true anomaly.

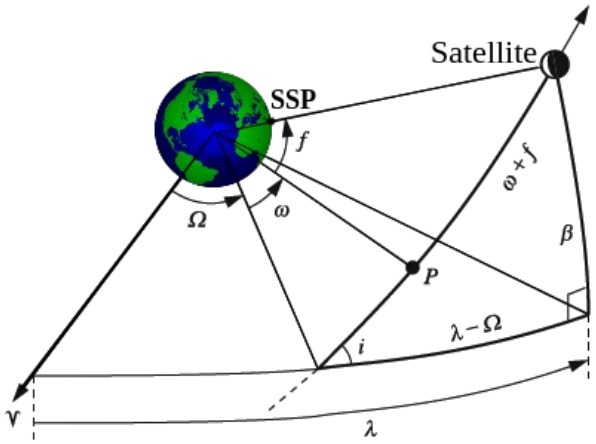


Figure 1: Geometric diagram illustrating SSP perturbations due to orbital inclination [2]

The spherical triangle in Figure 1 is used, yielding the following relations:

$$\frac{\sin\left(\frac{\pi}{2}\right)}{\sin(i)} = \frac{\sin(\beta)}{\sin(\omega + f)} \tag{1}$$

$$\cos\left(\frac{\pi}{2}\right)\sin(\beta) = -\cos(i)\sin(\omega + f)\cos(\lambda - \Omega) + \cos(\omega + f)\sin(\lambda - \Omega) \tag{2}$$

which simplifies to

$$\tan(\lambda - \Omega) = \cos(i)\tan(\omega + f) \tag{3}$$

where f is the true anomaly.

Assuming the spacecraft crosses the ascending node at $t = 0$, $\omega + f = \omega_0 t$, and:

$$\sin(\delta_{lat}) = \sin(i)\sin(\omega_0 t) \tag{4}$$

$$\cos(i) = \tan(\omega_0 t + \delta_{lon})\cot(\omega_0 t) \tag{5}$$

For small i, δ_{lat}, and δ_{lon}, the expressions reduce to:

$$\delta_{lat} = i\sin(\omega_0 t) \tag{6}$$

$$\delta_{lon} = -\frac{i^2}{4}\sin(2\omega_0 t) \tag{7}$$

2.2 Eccentricity-Induced Perturbations

By applying Kepler's second law, the satellite's angular velocity is maximum at perigee, minimum at apogee, and equal to Earth's angular velocity at points A and B (i.e., $E = \frac{\pi}{2}$ and $E = \frac{3\pi}{2}$), indicating no deviation. At both apogee and perigee, the sub-satellite point is at the center. At A and B, the sub-satellite point is furthest east and west, respectively [1].

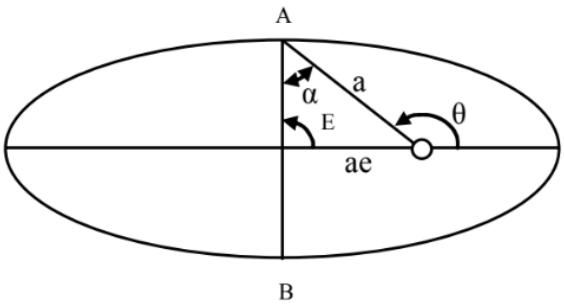


Figure 2: Geometric diagram illustrating SSP perturbations due to orbital eccentricity [1]

The longitude drift due to eccentricity is given by a sinusoid with an amplitude $\Delta\lambda$, which can be calculated at point A ($E = \frac{\pi}{2}$) where it is maximum:

$$\Delta\lambda = \theta - M\Big|_{E=\frac{\pi}{2}} \tag{8}$$

Using the formulation of the mean anomaly M:

$$M = E - e\sin E\Big|_{E=\frac{\pi}{2}} = \frac{\pi}{2} - e \tag{9}$$

Using elementary trigonometric relations:

$$\sin(\alpha) = \frac{ae}{a} = e \tag{10}$$

For small e and α, equation (10) can be simplified to

$$\alpha \approx e \tag{11}$$

By geometry,

$$\theta = \frac{\pi}{2} + \alpha = \frac{\pi}{2} + e \tag{12}$$

Thus, the drift in longitude is:

$$\Delta\lambda = 2e \tag{13}$$

Therefore, eccentricity in the circular orbit causes an east-west oscillation of the sub-satellite point with amplitude $2e$ [1].

2.3 Perturbations Due to Orbital Time-Period Deviations

For an ideal geosynchronous orbit with a period T equal to a sidereal day, the semi-major axis $\text{sma}_{\text{ideal}}$ is 42,164.16 km [3]. Let the spacecraft's semi-major axis be sma. The deviation is:

$$\Delta\text{sma} = \text{sma} - \text{sma}_{\text{ideal}} \tag{14}$$

Since $\text{sma} \neq \text{sma}_{\text{ideal}}$, the spacecraft's period is not equal to the Earth's rotation period. The angular velocity ω_0 is:

$$\omega_0 = \sqrt{\frac{\mu_E}{\text{sma}}} \quad \text{and} \quad \omega_{\text{ideal}} = \sqrt{\frac{\mu_E}{\text{sma}_{\text{ideal}}^3}} \tag{2.10}$$

This deviation from the ideal semi-major axis causes a shift in the Ascending Node over time, denoted by Δn. The shift is measured along the equator from one ascending node to the next, with the shift being positive in the direction of spacecraft motion. Thus, $360° - \Delta n$ represents the Earth's rotation during one orbit, as shown in Figure 3.

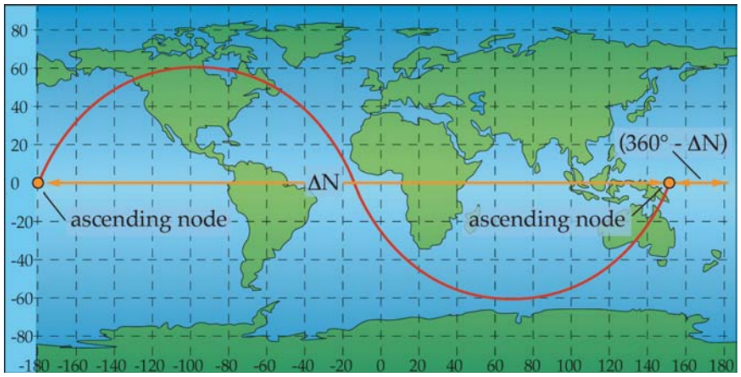


Figure 3: Ascending Node Shift Due to the Rotating Earth [3]

For spacecraft in a prograde orbit with period less than 24 hours :

$$\frac{360° - \Delta n}{\text{Period (hrs)}} = \frac{15°}{\text{hr}} \tag{15}$$

From the analytical result in Agrawal [1], the longitudinal deviation is:

$$\delta_{\text{lon}} = \Delta n = -0.013^\circ/\text{day} \times \frac{\Delta \text{sma}}{\text{km}} \tag{16}$$

NOTE: The deviations in orbital time-period are cumulative and require periodic corrections.

2.4 Graphical Representation of Latitude-Longitude Variations

The SSP's latitude and longitude variations, due to eccentricity, inclination, and non-ideal radius, are shown in Figures 4 to 11. The spacecraft is assumed to start at the ascending node.

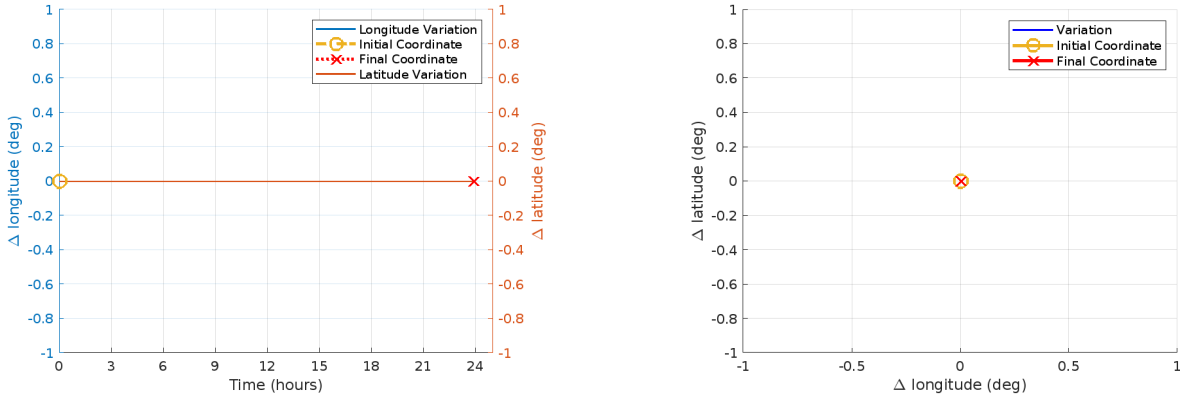


Figure 4: Latitude Longitude Variation for a = a_{ideal} , e = 0, i = 0°

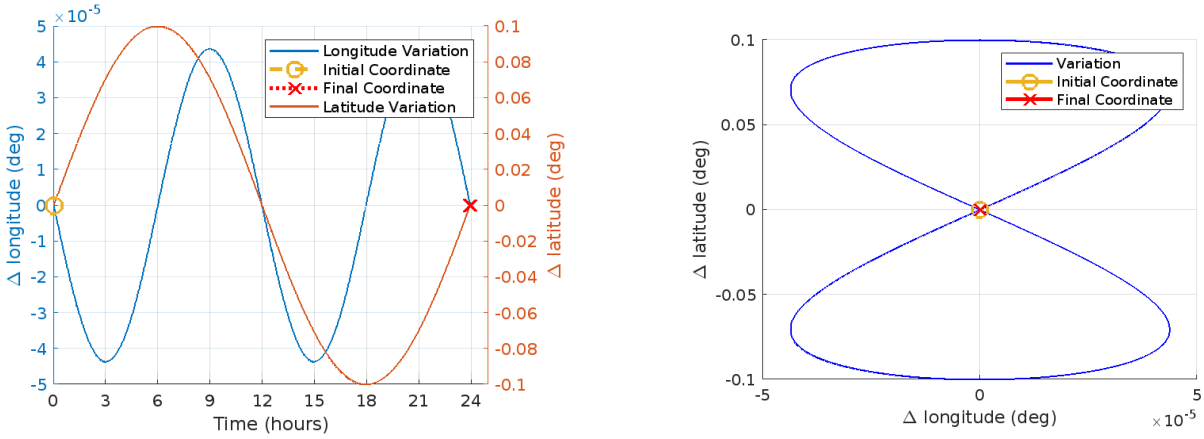


Figure 5: Latitude Longitude Variation for a = a_{ideal} , e = 0, i ≠ 0°

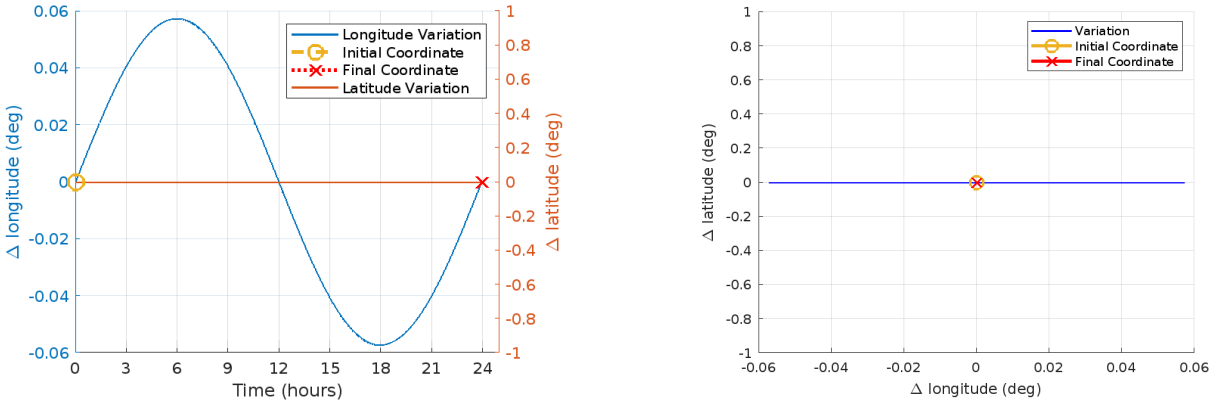


Figure 6: Latitude Longitude Variation for a = a_{ideal} , e $\neq$ 0, i = 0°

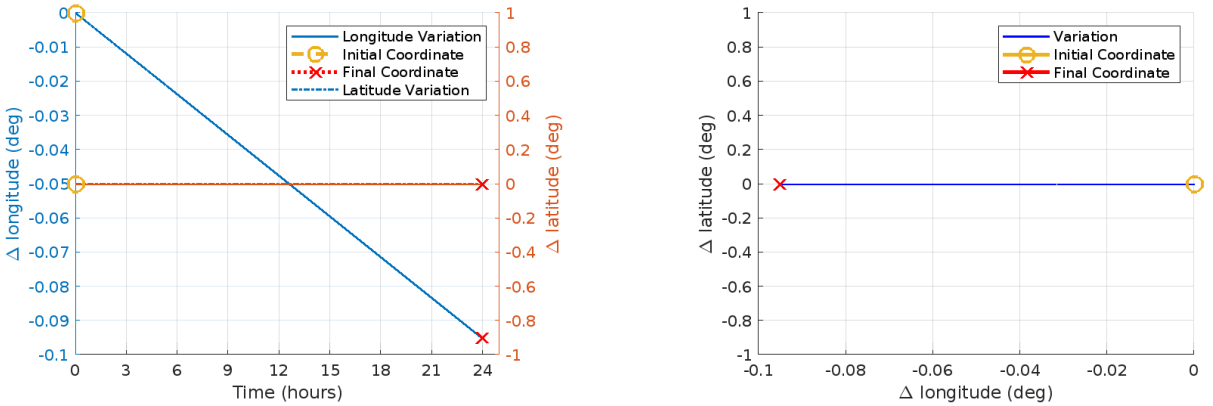


Figure 7: Latitude Longitude Variation for a $\neq$ a_{ideal} , e = 0, i = 0°

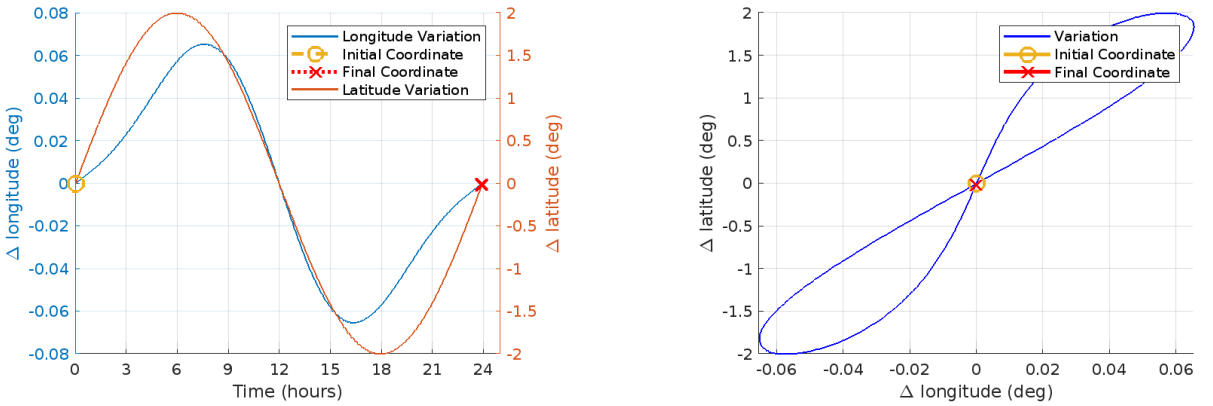


Figure 8: Latitude Longitude Variation for a = a_{ideal} , e $\neq$ 0, i $\neq$ 0°

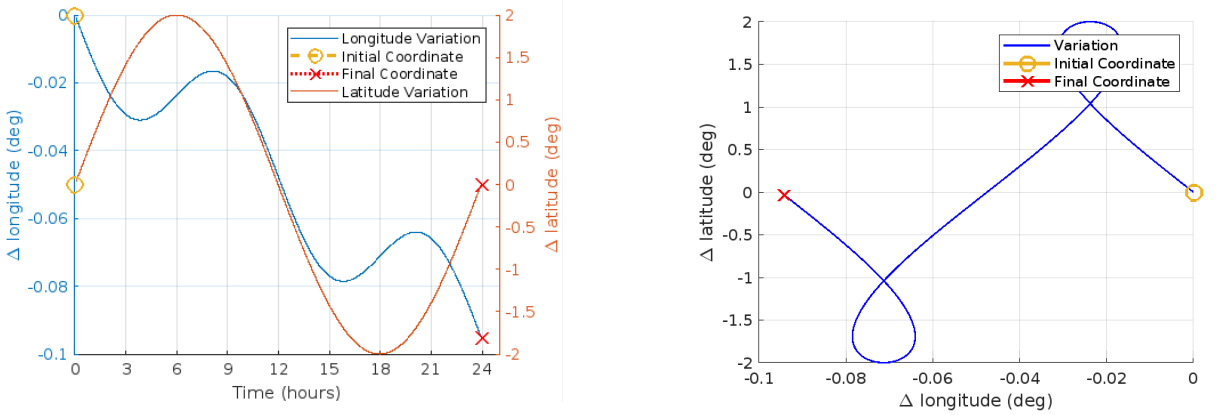


Figure 9: Latitude Longitude Variation for a $\neq$ a_{ideal} , e = 0, i $\neq$ 0°

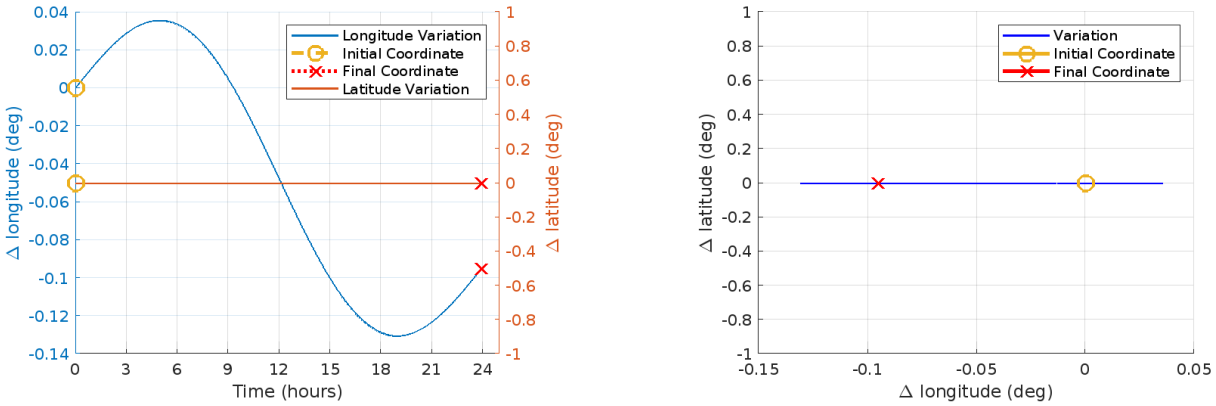


Figure 10: Latitude Longitude Variation for a $\neq$ a_{ideal} , e $\neq$ 0, i = 0°

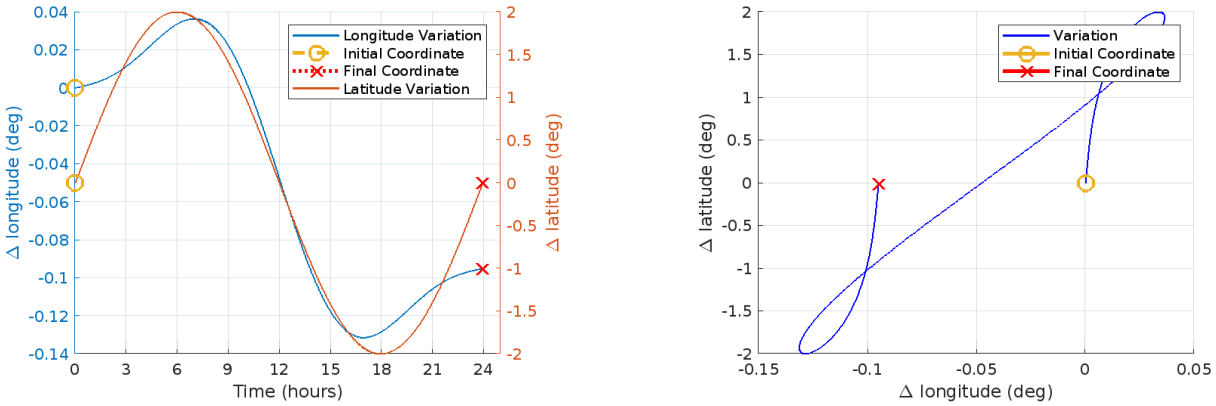


Figure 11: Latitude Longitude Variation for a $\neq$ a_{ideal} , e $\neq$ 0, i $\neq$ 0°

3 Geometric Characterization of Satellite-Earth Interactions

The geometric characterization of satellite-Earth interactions through precise calculation of Earth's angular radius and ground station positioning relative to the Sub-Satellite Point (SSP) is fundamental to comprehensive mission planning. These geometric parameters critically define coverage boundaries, ground station alignment, and payload pointing strategies by establishing the spatial relationship between Earth-based objects as viewed from the spacecraft. Calculating the angular radius enables precise determination of visible regions, minimizes signal degradation through atmospheric attenuation, bounds attitude maneuvers, and optimizes payload efficiency. The methodology provides a robust framework for transforming orbital mechanics principles into actionable pointing commands, ensuring accurate communication link establishment, target visibility assessment, and mission operational optimization under diverse orbital conditions.

3.1 Earth Angular Radius Calculation

The angular radius of the Earth as seen from a spacecraft involves computing ρ and λ_0, as shown in Figure 12, where:

- ρ: Angular radius of the spherical Earth from the spacecraft.
- λ_0: Angular radius of the observed region measured from Earth's center.

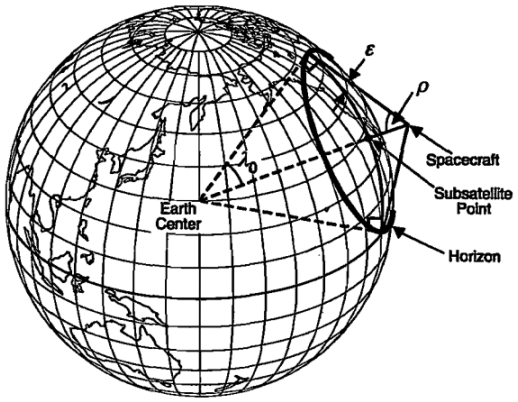


Figure 12: Geometry between the spacecraft and Earth's center [4].

Assuming a spherical Earth, the line from the spacecraft to the Earth's horizon is perpendicular to Earth's radius, giving:

$$\sin\rho = \cos\lambda_0 = \frac{R_E}{R_E + H}, \tag{17}$$

$$\rho + \lambda_0 = 90^\circ, \tag{18}$$

where R_E is Earth's radius and H is the satellite altitude.

The variation of Earth's angular radius from the satellite perspective with respect to altitude is shown in Figure 13.

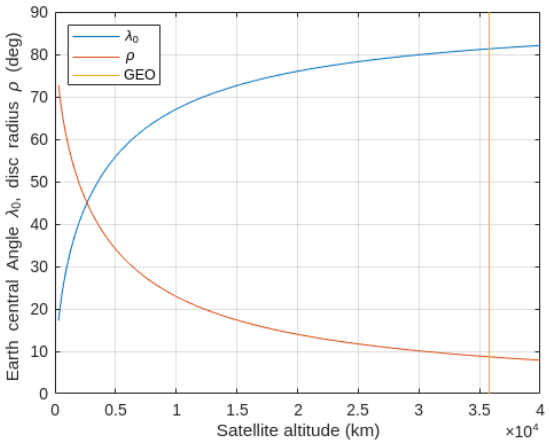


Figure 13: Variation of Earth Angular Radius with Satellite Altitude

3.2 Ground Station Position Relative to the Subsatellite Point

The position of the ground station relative to the subsatellite point (SSP) can be expressed in terms of spacecraft viewing angles.

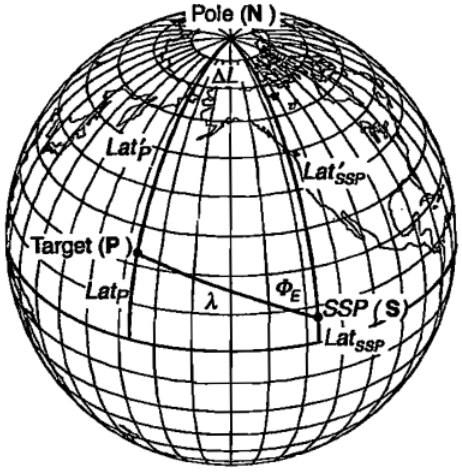


Figure 14: Geometric Relationship Between the Target and SSP on Earth's Surface [4]

From the trigonometric relationships of the spherical triangle ΔPNS in Figure 14, the azimuth angle ϕ_E, measured eastward from the north, and the angular distance λ between the subsatellite point (SSP) and the target (P) can be determined as follows:

$$\cos\lambda = \cos(Lat'_P)\cos(Lat'_{SSP}) + \sin(Lat'_P)\sin(Lat'_{SSP})\cos(\Delta L) \tag{19}$$

$$\cos(Lat'_P) = \cos\lambda\cos(Lat'_{SSP}) + \sin\lambda\sin(Lat'_{SSP})\cos(\phi_E) \tag{20}$$

Here, $Lat'_{SSP} = \pi/2 - Lat_{SSP}$, $Lat'_P = \pi/2 - Lat_P$, and $\Delta L = |Lon_{SSP} - Lon_P|$.

Simplifying these equations for practical use yields:

$$\cos\lambda = \sin(Lat_P)\sin(Lat_{SSP}) + \cos(Lat_P)\cos(Lat_{SSP})\cos(\Delta L) \tag{21}$$

$$\cos(\phi_E) = \frac{\sin(Lat_P) - \cos\lambda\sin(Lat_{SSP})}{\sin\lambda\cos(Lat_{SSP})} \tag{22}$$

The target P is east of the SSP if $\phi_E < 180°$ and west of the SSP if $\phi_E > 180°$.
By applying the sine rule:

$$\frac{\sin(\Delta L)}{\sin\lambda} = \frac{\sin(\phi_E)}{\sin(Lat'_P)} = \frac{\sin(\phi_E)}{\sin(\pi/2 - Lat_P)} = \frac{\sin(\phi_E)}{\cos(Lat_P)} \tag{23}$$

This results in:

$$\sin(\phi_E) = \frac{\sin(\Delta L)\cos(Lat_P)}{\sin(\lambda)} \tag{24}$$

Equations (22) and (24) then give:

$$\tan(\phi_E) = \frac{\sin(\Delta L)\cos(Lat_P)\cos(Lat_{SSP})}{\sin(Lat_P) - \cos(\lambda)\sin(Lat_{SSP})} \tag{25}$$

3.3 Implications of Spatial Dynamics on Mission Design

The geometric analysis of satellite-Earth interactions presented in this section establishes a critical computational framework that extends far beyond immediate mathematical abstraction. By precisely quantifying the angular relationships between spacecraft, Earth's surface, and ground stations, this study provides a fundamental understanding of the complex spatial dynamics inherent in orbital systems.

The angular radius calculations, represented by equation (17), offer profound insights into how spacecraft altitude fundamentally transforms Earth's observable geometry. Figure 13 illustrates the nonlinear relationship between satellite altitude and Earth's angular radius, revealing critical implications for mission planning. At lower altitudes, the angular radius increases rapidly, significantly expanding visible surface areas and communication potential. Conversely, at higher altitudes, the rate of change becomes more gradual, directly influencing payload design, communication strategies, and overall mission architecture.

By precisely calculating azimuth angles, angular distances, and relative positioning, mission designers gain unprecedented insights into the geometric constraints and opportunities presented by different orbital configurations. From a mission design perspective, these geometric characterizations serve as a critical foundation for subsequent analytical and simulation stages. While the immediate pointing command generation and control system design may employ different mathematical formulations, the fundamental geometric understanding established here underpins the entire mission development process. The precise quantification of visibility envelopes, grazing angles, and spatial relationships provides essential contextual information that informs design decisions across multiple mission domains.

Signal propagation analysis represents another crucial domain where these geometric insights prove invaluable. The detailed understanding of spacecraft-Earth geometry enables more sophisticated predictions of potential communication challenges, atmospheric attenuation effects, and electromagnetic propagation characteristics. Although not directly implemented in subsequent sections, these geometric principles provide a critical theoretical framework for understanding the fundamental constraints and opportunities of satellite communication.

The computational approach presented here transcends traditional geometric analysis by providing a unified framework that integrates orbital mechanics, spatial positioning, and mission design considerations. By transforming complex spatial relationships into tractable mathematical models, this study establishes a rigorous foundation for understanding the intricate interactions between spacecraft and Earth. The equations and methodologies introduced serve as a critical reference point for comprehensive mission planning, offering insights that extend beyond immediate computational applications.

As the subsequent sections progress to detailed control system design and pointing command generation, the geometric characterizations developed here will implicitly inform and constrain the analytical approach. The mathematical models presented establish a fundamental understanding of the spatial dynamics that underlie the entire mission design process, providing a robust theoretical framework for advanced spacecraft mission planning.

4 Attitude Command Formulation for Ground Station Pointing

Precise ground station pointing is critical for maintaining reliable communication with satellites, ensuring efficient data transmission and command reception. This requires effective attitude control strategies, factoring in the spacecraft's altitude, which influences coverage, grazing angles, and the Earth's subtended angle. These parameters define the bounds for attitude maneuvers. This section presents techniques for generating attitude commands to achieve stable pointing, with formulations validated for ISRO's TTC Ground Station at Lucknow under varying orbital conditions. The roll and yaw compensation commands are calculated using the latitude and longitude deviation of the SSP from the Ground Station Location, using figure 15 as reference.

4.1 Formulation Aligning with Attitude Commands by Bang [5]

From Figure 15, the distance D can be expressed in terms of R_E, H, and λ as:

$$D^2 = (R_E + H)^2 + R_E^2 - 2(R_E + H)R_E \cos\lambda \tag{26}$$

Substituting this into the sine rule:

$$\frac{R_E}{D} = \frac{\sin\eta}{\sin\lambda} \tag{27}$$

yields the relationship:

$$\sin\eta = \frac{R_E \sin\lambda}{\sqrt{(R_E + H)^2 + R_E^2 - 2(R_E + H)R_E \cos\lambda}} \tag{28}$$

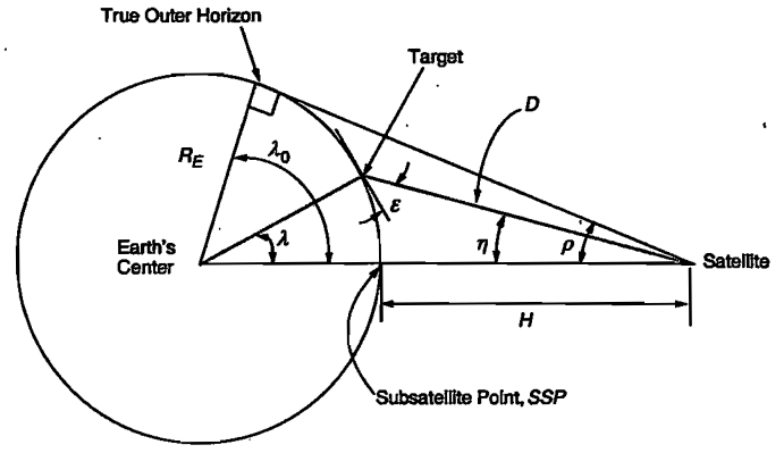


Figure 15: Angular Relationships Between Satellite, Target, and Earth Center [4]

From orbital definitions, illustrated in Figure 16, and the results in section 2.4, the perturbation in latitude is observed to be significantly larger than in longitude, both being much smaller than 1 radian. Considering the magnitudes of these perturbations and the use of a 3-2-1 rotation sequence, it can be approximated that:

- Positive pitch induces a change in the SSP longitude along the East.
- Positive roll induces a change in the SSP latitude along the North.

Consequently, η corresponds to roll for $\lambda = \Delta\text{Lat}$, and to pitch for $\lambda = \Delta\text{Lon}$ in Equation (28). Thus, the relationship between latitude-longitude deviations and spacecraft attitude can be derived using Equation (28) and Figure 17.

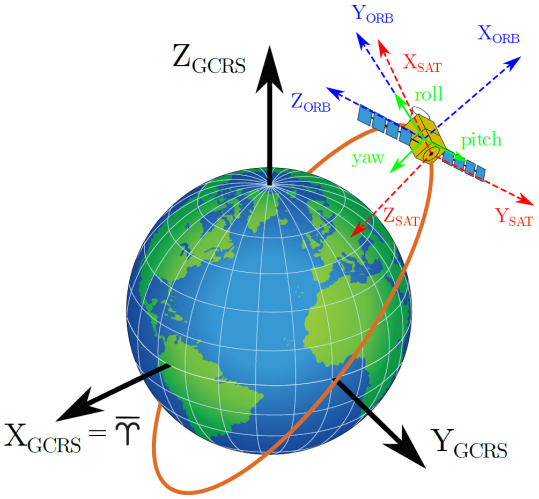


Figure 16: Definition of the Spacecraft Co-ordinate System [6]

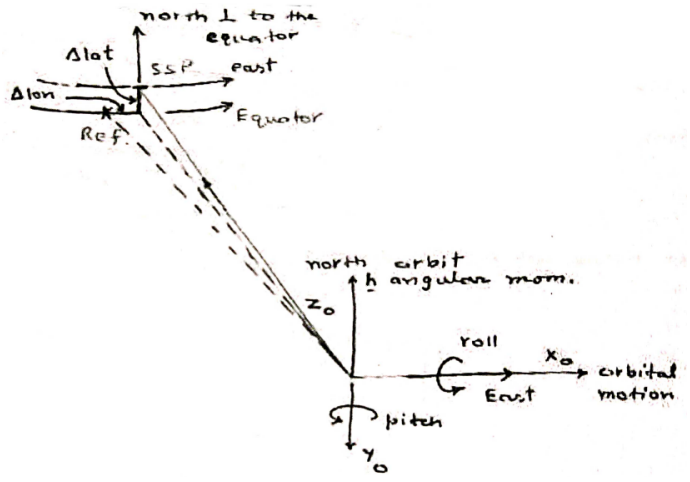


Figure 17: Relationship between Latitude-Longitude Deviation and Spacecraft Attitude

4.2 Formulation by elimination D

Another approach to derive the relation between η and λ to obtain pointing attitude commands is by eliminating D and capturing it in terms of ρ

$$\frac{R_E}{\sin(\eta)} = \frac{R_E + H}{\sin(90^\circ + \epsilon)} = \frac{D}{\sin(\lambda)} \tag{29}$$

using the equality of sum of internal angles of a triangle,

$$\eta + \lambda + \epsilon = 90^\circ \tag{30}$$

equation (29) gives

$$\frac{R_E}{R_E + H} = \frac{\sin(\eta)}{\sin(\lambda + \eta)} \tag{31}$$

The ratio of distances in equation (31) can be related to ρ and thus, the equation reduces to

$$\frac{\sin(\eta)}{\sin(\lambda + \eta)} = \sin(\rho) \tag{32}$$

which simplifies to

$$\sin(\eta) = \sin(\rho)\sin(\lambda + \eta) \tag{33}$$

$$= \sin(\rho)\left[\sin(\lambda)\cos(\eta) + \cos(\lambda)\sin(\eta)\right] \tag{34}$$

simplifying the above equation yields

$$\tan(\eta) = \frac{\sin(\rho)\sin(\lambda)}{1 - \sin(\rho)\cos(\lambda)} \tag{35}$$

Similar to the above reasoning, the roll-pitch commands can be obtained as

$$\phi = -\tan^{-1}\left[\frac{\sin(\rho)\sin(\Delta Lat)}{1 - \sin(\rho)\cos(\Delta Lat)}\right] \tag{36}$$

$$\theta = -\tan^{-1}\left[\frac{\sin(\rho)\sin(\Delta Lon)}{1 - \sin(\rho)\cos(\Delta Lon)}\right] \tag{37}$$

4.3 Pitch and Yaw Pointing in Satellites: Relevance and Scope

Pitch control in bias momentum satellites maintains Earth-pointing attitude and compensates for orbital deviations. Pitch adjustments compensate for variations in the geosynchronous orbit's semi-major axis length, which influence the Right Ascension of the Ascending Node (RAAN) drift.

Yaw orientation optimizes solar panel alignment for power generation, ensures proper radiator exposure for thermal control, and maintains consistent illumination for Earth observation.

In bias momentum satellites, while roll and yaw dynamics are coupled due to the gyroscopic effect of the momentum wheel, the pitch axis remains independently controllable. This decoupling of pitch dynamics from roll-yaw behavior enables independent RAAN drift management without affecting other orbital elements.

This work primarily analyzes coupled roll-yaw dynamics, leveraging pitch independence and assuming zero-degree nominal yaw pointing.

4.4 Attitude Commands for Pointing at ISTRAC Ground Station, Lucknow

Figures 18 to 21 display the pointing commands generated for the ISTRAC Ground Station in Lucknow (26.92°N, 80.95°E). These commands are based on the orbital parameters outlined in Table 1, which includes the configurations for selected INSAT, GSAT, and IRNSS satellites. These satellites were specifically chosen for their proximity to the 81°E longitude, where the ISTRAC Ground Station is located, ensuring relevance to the station's operational coverage.

Satellite	Inclination	Eccentricity	RAAN	Application
INSAT-3DR	0.1°	0.0003	74°E	Weather monitoring
GSAT-7	1.0°	0.0003	74°E	Maritime Defense Communications
GSAT-10	0.1°	0.0002	83°E	Telecommunications
IRNSS-1G	28.1°	0.0012	55°E	NavIC; Regional navigation

Table 1: Orbital Parameters of Selected INSAT, GSAT, and IRNSS Satellites

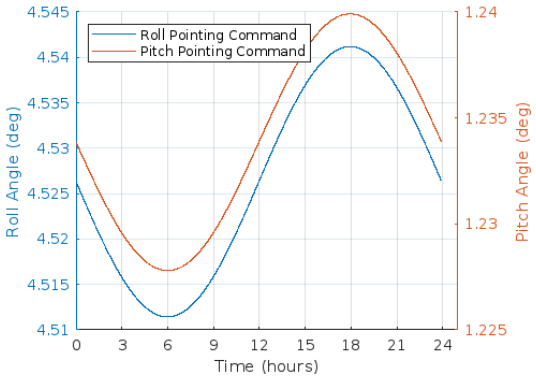


Figure 18: INSAT-3DR

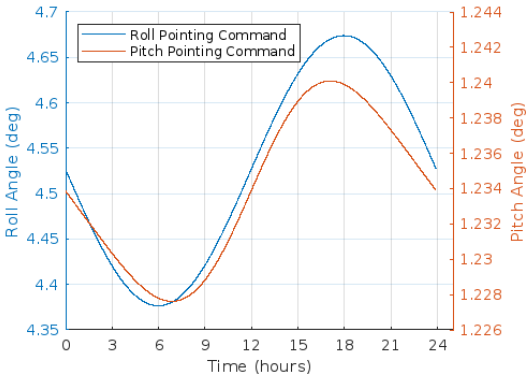


Figure 19: GSAT-7

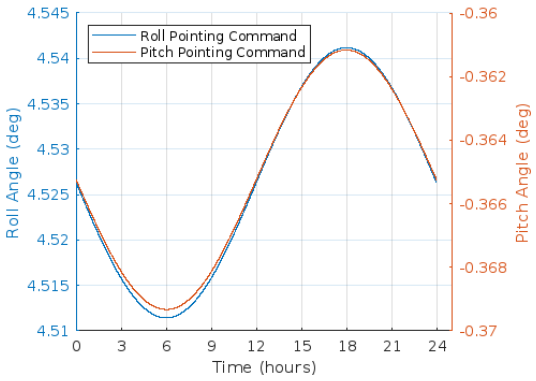


Figure 20: GSAT-10

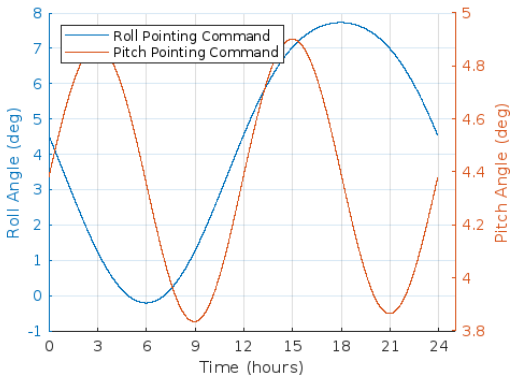


Figure 21: IRNSS-1G

In low-inclination orbital configurations, satellite attitude control presents unique challenges, particularly with respect to roll and pitch pointing dynamics. The roll pointing requirements become substantially more significant compared to pitch pointing, creating a critical design consideration for attitude control systems. Satellites such as the INSAT-3DR, GSAT-7, and the GSAT-10, which operate in near-equatorial orbits, represent ideal candidates for this configuration. Their orbital characteristics create pointing scenarios where a roll-actuated offset thruster can efficiently maintain pitch accuracy while allowing for effective roll adjustments.

The lower the orbital inclination, the more pronounced the roll dynamics become, necessitating a sophisticated yet simplified attitude control strategy. By leveraging the natural coupling between roll and pitch errors, which interchange approximately every quarter of the orbital period, the bias momentum system can effectively manage attitude control with reduced complexity. The offset thruster configuration enables simultaneous torque generation about multiple axes, allowing precise roll control while maintaining sufficient pitch accuracy. This approach is particularly advantageous for satellites operating in near-equatorial orbits, where traditional three-axis stabilization methods would require more complex and resource-intensive control mechanisms. The inherent ability to attenuate pitch errors through roll angle measurements provides a robust and energy-efficient solution for attitude management in low-inclination orbital scenarios.

5 Attitude Dynamics of Bias Momentum Satellites

A bias momentum satellite, or momentum-biased satellite, uses a momentum wheel for primary attitude control and stabilization. Analyzing these dynamics helps in modeling and controlling the spacecraft's attitude, ensuring stability and alignment with its intended direction. This analysis also aids in identifying and compensating for external disturbances and developing control algorithms to manage attitude, mitigate disturbances, and maintain accurate pointing.

5.1 Co-ordinate Frames

The development of spacecraft dynamics begins with defining coordinate frames and transformations between them. A minimum of three frames is essential:

- **Earth-Centered Inertial (ECI) Frame**: Used for calculating the latitude and longitude of the spacecraft's center of mass along its orbit.
- **Local Rotating or Orbiting Frame**: Originating at the spacecraft's center of mass, it rotates around the orbital plane. The z-axis points toward the Earth's center, and the x-axis aligns with the direction of motion.
- **Body Frame**: Attached to the spacecraft's center of mass and aligned with its principal axes. It moves and rotates with the spacecraft, capturing its attitude and orientation using the standard 3-2-1 rotation sequence.

These frames, illustrated in Figures 22 and 23, provide the foundation for modeling spacecraft dynamics and attitude control.

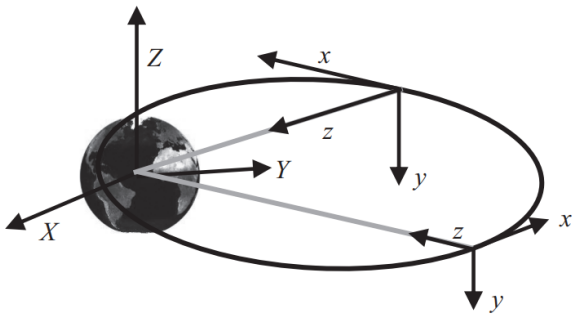


Figure 22: Definition of the inertial and orbital frames [7]

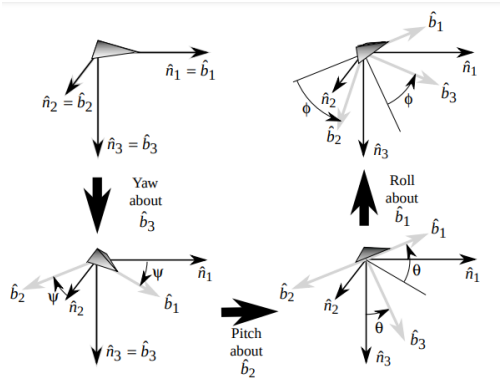


Figure 23: A (3-2-1) rotation sequence commonly used to describe the orientation of aerospace vehicles [8]

5.2 Governing Equations of Motion

The angular velocity in the inertial frame, expressed in the body coordinate system, is:

$$\vec{\omega_B} = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \tag{38}$$

The dynamics of momentum bias satellites involve coupling between momentum components due to inherent angular momentum provided by a spinning momentum wheel, oriented normal to the orbit plane [9]. This configuration enables roll-yaw coupling, allowing yaw control with roll measurements. The total angular momentum is:

$$\vec{H} = \begin{bmatrix} I_x\omega_x \\ I_y\omega_y \\ I_z\omega_z \end{bmatrix} + \begin{bmatrix} 0 \\ h_w \\ 0 \end{bmatrix} \tag{39}$$

where I_x, I_y, I_z are principal moments of inertia, and $h_w = -h$ with $h > 0$.

The Euler equations of motion are:

$$T_{cx} + T_{dx} + T_{gx} = I_x\dot{\omega}_x - (I_y - I_z)\omega_y\omega_z - h_w\omega_z \tag{40}$$
$$T_{cy} + T_{dy} + T_{gy} = I_y\dot{\omega}_y - (I_z - I_x)\omega_z\omega_x \tag{41}$$
$$T_{cz} + T_{dz} + T_{gz} = I_z\dot{\omega}_z - (I_x - I_y)\omega_x\omega_y + h_w\omega_x \tag{42}$$

where T_c, T_d, T_g are control, disturbance, and gravity gradient torques, respectively.

For a circular orbit, the gravity gradient torques simplify to:

$$T_{gx} = -3\omega_0^2(I_y - I_z)\phi \tag{43}$$
$$T_{gy} = -3\omega_0^2(I_x - I_z)\theta \tag{44}$$
$$T_{gz} \approx 0 \tag{45}$$

where ω_0 is the orbital angular velocity.

The body rates, related to Euler angle rates via a 3-2-1 transformation, are:

$$\begin{bmatrix}\dot{\phi}\\ \dot{\theta}\\ \dot{\psi}\end{bmatrix} = \begin{bmatrix}1 & 0 & \theta\\ 0 & 1 & -\phi\\ 0 & \phi & 1\end{bmatrix}\begin{bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{bmatrix} + \omega_0\begin{bmatrix}\psi\\ 1\\ 0\end{bmatrix} \tag{46}$$

Expressing the body rates in terms of the euler angle rates from equation (46), we get,

$$\begin{bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{bmatrix} = \begin{bmatrix}1 & 0 & -\theta\\ 0 & 1 & \phi\\ 0 & -\phi & 1\end{bmatrix}\begin{bmatrix}\dot{\phi}\\ \dot{\theta}\\ \dot{\psi}\end{bmatrix} - \omega_0\begin{bmatrix}1 & 0 & -\theta\\ 0 & 1 & \phi\\ 0 & -\phi & 1\end{bmatrix}\begin{bmatrix}\psi\\ 1\\ 0\end{bmatrix} \tag{47}$$

Assuming small Euler angles and neglecting higher-order terms [9], the equations further simplify for linearized analysis. Equation (47) simplifies to give the linearized body rates as,

$$\begin{bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{bmatrix} = \begin{bmatrix}\dot{\phi} - \theta\dot{\psi} - \omega_0\psi\\ \dot{\theta} + \phi\dot{\psi} - \omega_0\\ -\phi\dot{\theta} + \dot{\psi} + \omega_0\psi\end{bmatrix} = \begin{bmatrix}\dot{\phi} - \omega_0\psi\\ \dot{\theta} - \omega_0\\ \dot{\psi} + \omega_0\phi\end{bmatrix} \tag{48}$$

and the linearized body rate derivatives as,

$$\begin{bmatrix}\dot{\omega_x}\\ \dot{\omega_y}\\ \dot{\omega_z}\end{bmatrix} = \begin{bmatrix}\ddot{\phi} - \dot{\theta}\dot{\psi} - \theta\ddot{\psi} - \omega_0\dot{\psi}\\ \ddot{\theta} + \dot{\phi}\dot{\psi} + \phi\ddot{\psi}\\ \ddot{\psi} - \dot{\phi}\dot{\theta} - \psi\ddot{\theta} + \omega_0\dot{\phi}\end{bmatrix} = \begin{bmatrix}\ddot{\phi} - \omega_0\dot{\psi}\\ \ddot{\theta}\\ \ddot{\psi} + \omega_0\dot{\phi}\end{bmatrix} \tag{49}$$

Substituting the equations (48), (49) and (43) to (45) in the equations of motion (40) to (42), we get the equations of system dynamics as

$$T_{dx} + T_{cx} = I_x\ddot{\phi} + (a - \omega_0 h_w)\phi + (b - h_w)\dot{\psi} \tag{50}$$

$$T_{dy} + T_{cy} = I_y\ddot{\theta} + d\,\theta + \dot{h} \tag{51}$$

$$T_{dz} + T_{cz} = I_z\ddot{\psi} + (c - \omega_0 h_w)\psi - (b - h_w)\dot{\phi} \tag{52}$$

where : $a = 4{\omega_0}^2(I_y - I_z)$; $b = -\omega_0(I_x - I_y + I_z)$; $c = {\omega_0}^2(I_y - I_x)$; $d = 3{\omega_0}^2(I_x - I_z)$

5.3 Time Domain Analysis

The derived equations of motion can be reformulated into a **State-Space Representation**, a crucial framework for implementing control systems. Focusing on the coupled roll and yaw dynamics, the second-order ordinary differential equations (ODEs) are reduced to first-order form by introducing the following state variables:

$$\phi = x_1, \tag{53}$$

$$\dot{\phi} = x_2, \tag{54}$$

$$\psi = x_3, \tag{55}$$

$$\dot{\psi} = x_4. \tag{56}$$

Here, ϕ and ψ represent the roll and yaw angles, respectively, while $\dot{\phi}$ and $\dot{\psi}$ represent their angular velocities. Using these definitions, the system of ODEs becomes:

$$\dot{x}_1 = x_2, \tag{57}$$

$$\dot{x}_2 = \frac{T_{cx} + T_{dx}}{I_x} - \frac{(a + \omega_0 h)}{I_x} x_1 - \frac{(b + h)}{I_x} x_4, \tag{58}$$

$$\dot{x}_3 = x_4, \tag{59}$$

$$\dot{x}_4 = \frac{T_{cz} + T_{dz}}{I_z} - \frac{(c + \omega_0 h)}{I_z} x_3 + \frac{(b + h)}{I_z} x_2. \tag{60}$$

These equations capture the evolution of the roll and yaw dynamics in terms of the state variables x_1, x_2, x_3, and x_4, where T_{cx} and T_{cz} are control torques, and T_{dx} and T_{dz} represent disturbances.

To simplify and generalize these equations for control design, they are expressed in the **State-Space Representation** as:

$$\dot{\vec{\mathbf{x}}} = \mathbf{A}\vec{\mathbf{x}} + \mathbf{B}(\vec{\mathbf{u}} + \vec{\mathbf{d}}), \tag{61}$$

where the state vector, input vector, and disturbance vector are defined as:

$$\vec{\mathbf{x}} = \begin{bmatrix} \phi \\ \dot{\phi} \\ \psi \\ \dot{\psi} \end{bmatrix}, \quad \vec{\mathbf{u}} = \begin{bmatrix} T_{cx} \\ T_{cz} \end{bmatrix}, \quad \vec{\mathbf{d}} = \begin{bmatrix} T_{dx} \\ T_{dz} \end{bmatrix} \tag{62}$$

The matrices $\mathbf{A}$ and $\mathbf{B}$, which define the system dynamics and input coupling, respectively, are :

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{-(a+\omega_0 h)}{I_x} & 0 & 0 & \frac{-(b+h)}{I_x} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{(b+h)}{I_z} & \frac{-(c+\omega_0 h)}{I_z} & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ \frac{1}{I_x} & 0 \\ 0 & 0 \\ 0 & \frac{1}{I_z} \end{bmatrix} \tag{63}$$

This State-Space Representation provides a structured way to model the system, making it highly useful for the future implementation of optimal control techniques.

5.4 Frequency Domain Analysis

5.4.1 System Response Frequency

The Laplace transform of the coupled ODEs (40) to (42) is obtained as

$$\begin{bmatrix} I_x s^2 + a + \omega_0 h & 0 & (b+h)s \\ 0 & I_y s^2 + d & 0 \\ -(b+h)s & 0 & I_z s^2 + c + \omega_0 h \end{bmatrix} \begin{bmatrix} \phi(s) \\ \theta(s) \\ \psi(s) \end{bmatrix} = \begin{bmatrix} I_x(\dot{\phi}_0 + s\phi_0) + (b+h)\psi_0 \\ I_x(\dot{\theta}_0 + s\theta_0) \\ I_z(\dot{\psi}_0 + s\psi_0) - (b+h)\phi_0 \end{bmatrix} + \begin{bmatrix} T_{cx} \\ T_{cy} \\ T_{cz} \end{bmatrix} + \begin{bmatrix} T_{dx} \\ T_{dy} \\ T_{dz} \end{bmatrix} \tag{64}$$

where $\phi_0, \dot{\phi}_0, \theta_0, \dot{\theta}_0, \psi_0, \dot{\psi}_0$ are initial conditions.

The analysis shows that the pitch dynamics are independent of the wheel and are not coupled with the roll and yaw. Therefore, only the coupled roll and yaw equations are considered in the subsequent analysis.

Approximation for high h

The angular momentum $\vec{\mathbf{h}}$ is chosen to greatly increase the coupling of the roll and yaw dynamics. [10] This leads to the relation

$$|h| \gg max[|a/\omega_0|, |b|, |c/\omega_0|, |\omega_0(I_x + I_z)|] \tag{65}$$

Equation (65) verified to hold for our simulation as well

$$\begin{aligned} h &= 20 \\ a/\omega_0 &\approx 0.18864 \\ b &= 0.15478 \\ c/\omega_0 &\approx 0.043 \\ \omega_0(I_x + I_z) &= 0.22 \end{aligned}$$

For the approximation mentioned in equation (65), the equations describing the roll and yaw dynamics reduce to

$$\begin{bmatrix} I_x s^2 + \omega_0 h & hs \\ -hs & I_z s^2 + \omega_0 h \end{bmatrix} \begin{bmatrix} \phi(s) \\ \psi(s) \end{bmatrix} = \begin{bmatrix} I_x(\dot{\phi_0} + s\phi_0) + h\psi_0 \\ I_z(\dot{\psi_0} + s\psi_0) - h\phi_0 \end{bmatrix} + \begin{bmatrix} T_{cx} \\ T_{cz} \end{bmatrix} + \begin{bmatrix} T_{dx} \\ T_{dz} \end{bmatrix} \tag{66}$$

Thus the characteristic equation for the roll-yaw response of the system is given by

$$\begin{vmatrix} \Delta(s) = I_x s^2 + \omega_0 h & hs \\ -hs & I_z s^2 + \omega_0 h \end{vmatrix} = I_x I_z s^4 + ((I_x + I_z)\omega_0 h + h^2)s^2 + (\omega_0 h)^2 \tag{67}$$

Equation (65) further reduces the characteristic equation to give

$$\Delta(s) = I_x I_z s^4 + h^2 s^2 + (\omega_0 h)^2 \tag{68}$$

The roots of the above characteristic equation are given by

$$s^2 = \frac{-h^2 \pm \sqrt{h^4 - 4{\omega_0}^2 h^2 I_x I_z}}{2I_x I_z} = \frac{-h^2 \pm h^2\sqrt{1 - \frac{4{\omega_0}^2 I_x I_z}{h^2}}}{2I_x I_z} \tag{69}$$

For our design parameters,

$$\frac{4\omega_0^2 I_x I_z}{h^2} \ll 1 \tag{70}$$

thus approximating the square root using the binomial expansion of $(1-x)^{\mathrm{n}}$,

$$(1+x)^{\mathrm{n}} = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3.... \tag{71}$$

Equation (70) is verified to hold for our simulation as well

$$\frac{4\omega_0^2 I_x I_z}{h^2} \approx 0.00012$$

Application of the binomial expansion gives

$$\sqrt{1-\frac{4{\omega_0}^2 I_x I_z}{h^2}} \approx 1-\frac{1}{2}\frac{4{\omega_0}^2 I_x I_z}{h^2}-\frac{1}{8}(\frac{4{\omega_0}^2 I_x I_z}{h^2})^2-.... \tag{72}$$

and hence

$$h^2\sqrt{1-\frac{4{\omega_0}^2 I_x I_z}{h^2}} \approx h^2-\frac{1}{2}4{\omega_0}^2 I_x I_z-\frac{1}{8}\frac{(4{\omega_0}^2 I_x I_z)^2}{h^2}-.... \tag{73}$$

Thus one pair of roots of the characteristic equation which correspond to the poles of the system are given as

$${s_{1,2}}^2=\frac{-\frac{1}{2}4{\omega_0}^2 I_x I_z-\frac{1}{8}\frac{(4{\omega_0}^2 I_x I_z)^2}{h^2}-....}{2I_x I_z}\approx -{\omega_0}^2 \quad \Longrightarrow \quad s_{1,2}=\pm j\omega_0 \tag{74}$$

similarly, the other pair is given by

$${s_{3,4}}^2=\frac{-2h^2+\frac{1}{2}4{\omega_0}^2 I_x I_z+\frac{1}{8}\frac{(4{\omega_0}^2 I_x I_z)^2}{h^2}-....}{2I_x I_z}\approx \frac{-h^2}{I_x I_z} \quad \Longrightarrow \quad s_{3,4}=\pm\frac{jh}{\sqrt{I_x I_z}} \tag{75}$$

The higher order terms are negligible with respect to the first term, and hence the proceeding higher order terms can be neglected in the approximation of the response frequencies.

The characteristic equation for the roll-yaw response has two pairs of roots on the imaginary axis. The values of these roots are approximately the orbit rate ω_0 and the nutation frequency $h/\sqrt{I_x I_z}$. The first pair of roots, $s_{1,2}$, reflect the fact that the reference axes are rotating about the earth at rate ω_0 and hence correspond to the Orbital Rate Poles of the system. The latter pair, $s_{3,4}$, are the nutation frequency which correspond to the Nutation Frequency Poles of the system.

Thus the characteristic equation can be simplified to

$$\Delta(s)=I_x I_z(s^2+\frac{h^2}{I_x I_z})(s^2+{\omega_0}^2) \tag{76}$$

Approximation by Iwens

Iwens [11] approximates response as the linearized dynamic equations having 2 pairs of imaginary roots : $s_{1,2}$ at orbit rate (ω_0) and $s_{3,4}$ at the nutation frequency (Ω_0) [11]

$$s_{1,2}=\pm j\omega_0, \qquad\qquad s_{3,4}=\pm j\Omega_0 \tag{77}$$

where,

$$\Omega_0=\sqrt{\frac{[h-\omega_0(I_x-I_y)][h-\omega_0(I_z-I_y)]}{I_x\ I_z}} \tag{78}$$

5.4.2 Splitting of Poles

Without the assumption of equation (65), we analyze the splitting of the poles of the characteristic equation with increasing momentum bias. The assumption has been relaxed for the analysis since it would not hold for all values of h starting from 0. Without the high momentum bias approximation, the characteristic equation for the roll-yaw response of the system is then given by :

$$\Delta(s) = \begin{vmatrix} I_x s^2 + a + \omega_0 h & (b+h)s \\ -(b+h)s & I_z s^2 + c + \omega_0 h \end{vmatrix} \tag{79}$$

$$\Delta(s) = I_x I_z s^4 + (I_x(c+\omega_0 h) + I_z((a+\omega_0 h)) + (b+h)^2)s^2 + (a+\omega_0 h)(c+\omega_0 h) \tag{80}$$

The magnitude of the complex roots of the characteristic equation given by equation (80) corresponds to the frequency of response of the system.

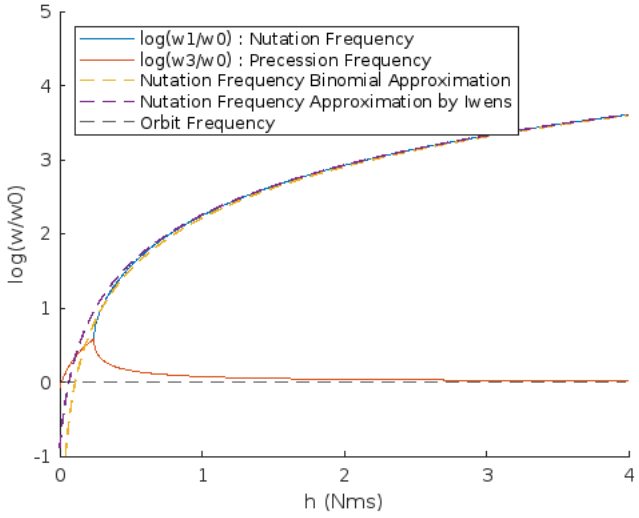


Figure 24: Splitting of system response poles due to momentum bias

The result shows how the momentum bias gives rise to a second set of response frequencies. It also verifies that the approximations hold for values of h $\geq$ 2,Nms

6 Controller Design and Implementation

The coupling of roll-yaw dynamics enables the stabilization of a satellite without the need for direct yaw measurements. As indicated by Sidi [9], this coupling allows sufficient control of the yaw axis when the control commands defined in Equation (52) is given by:

$$T_{cz} = a \; T_{cx} \tag{81}$$

The linear relationship between the control torques arises from the fact that roll and yaw errors interchange every quarter of an orbital period, as illustrated in Figure 25. Specifically, any accumulated yaw error transforms into a roll error after one-quarter of the orbit. Since roll angle measurements are available, the yaw error can be indirectly controlled by addressing the roll error,

which is subsequently attenuated. Consequently, a controller is employed to stabilize and achieve the desired roll angle, leveraging the availability of roll measurements.

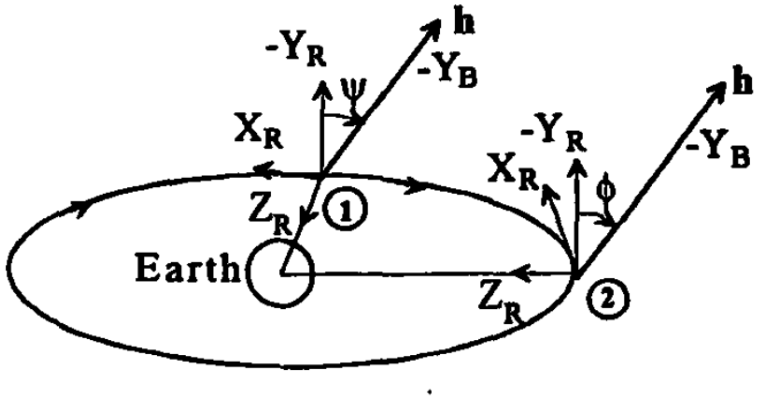


Figure 25: Roll-Yaw Error Exchange [9]

This linear relationship between control torques about the x-axis and z-axis enables the implementation of an offset thruster actuator system which comprises offset pairs of thrusters, as shown in Figure 26. This offset configuration allows control torques to be applied simultaneously about both the x- and z-axes. However, the effectiveness of this configuration depends on the proportionality between the control torques along these two axes. For bias-momentum satellites, offset thrusters provide a practical solution for simultaneous actuation.

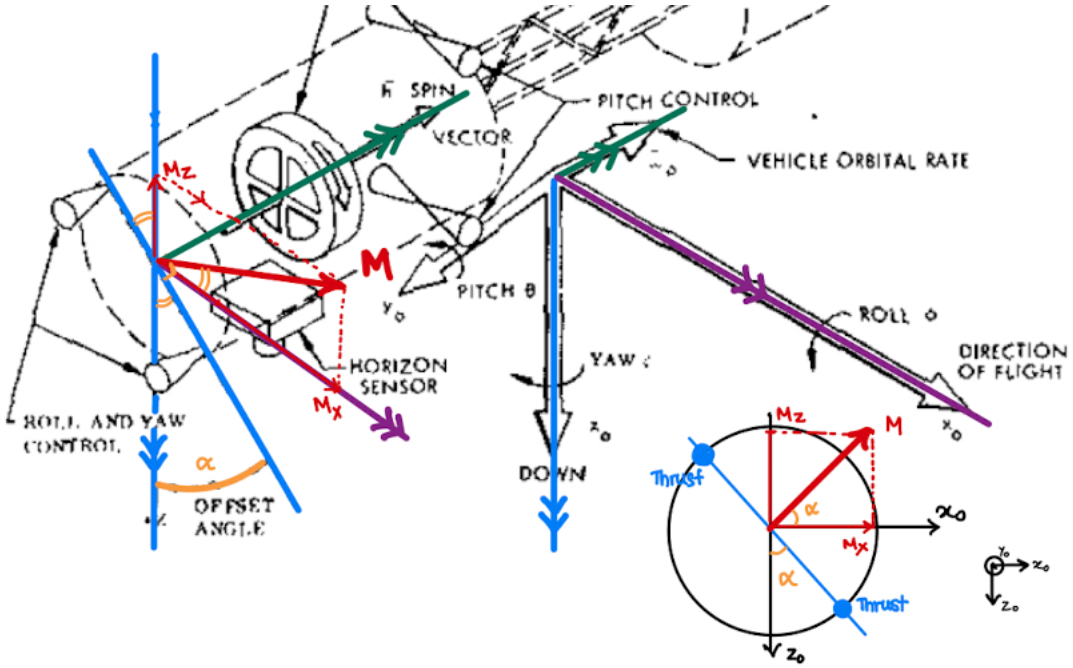


Figure 26: Offset Actuators Configuration (adapted from Dougherty [10])

These coupled dynamics along with offset thruster implementation and control law are represented in Iwens [11]. The closed loop roll channel can then be analysed to justify the nature of controller used to perform the task. The open loop roll transfer function considering offset thruster implementationn is given as

$$G(s)H_{roll}(s) = G(s)\cos\alpha\,\frac{I_z s^2 + h\,\tan\alpha\,s + \omega_0 h}{I_x I_z(s^2 + \frac{h^2}{I_x I_z})(s^2 + \omega_0^2)} \tag{82}$$

where G(s) is the controller, and other symbols are the same as mentioned in Section 5.

6.1 Controller Design using Root Locus Analysis

System Root Locus

The root locus of H_{roll}, representing the system without a controller, is plotted to analyze the feasibility of controlling the system with only a proportional controller.

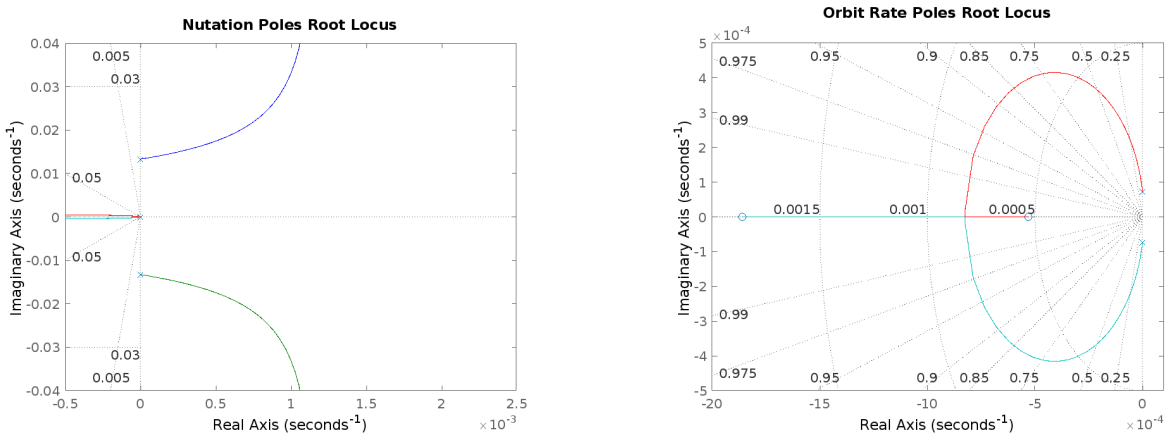


Figure 27: Root Locus of the System (Nutation and Orbit Poles)

Figure 27 shows the root locus of the system. Both subfigures are the same, with the second plot zoomed for clarity. The orbit rate poles are stabilized by the plant's zeros, ensuring BIBO stability at the orbital frequency. However, the nutation frequency poles shift to the right half-plane, indicating undamped behavior and a lack of BIBO stability at the nutation frequency. Thus, a proportional controller cannot achieve BIBO stability.

Addition of a Zero in the Right Half s-Plane

When a zero is added in the right half s-plane, the transfer function of the roll-error actuated controller becomes:

$$G(s) = (\tau s - 1) \tag{83}$$

This introduces a zero at $s = \frac{1}{\tau}$, where $\tau > 0$.

Figures 28 show the root locus with the added controller. Both subfigures represent the same root locus, zoomed for clarity. One system mode corresponding to the orbit rate pole moves toward the plant's zero, while the other moves toward the controller's zero. Since the controller zero is in

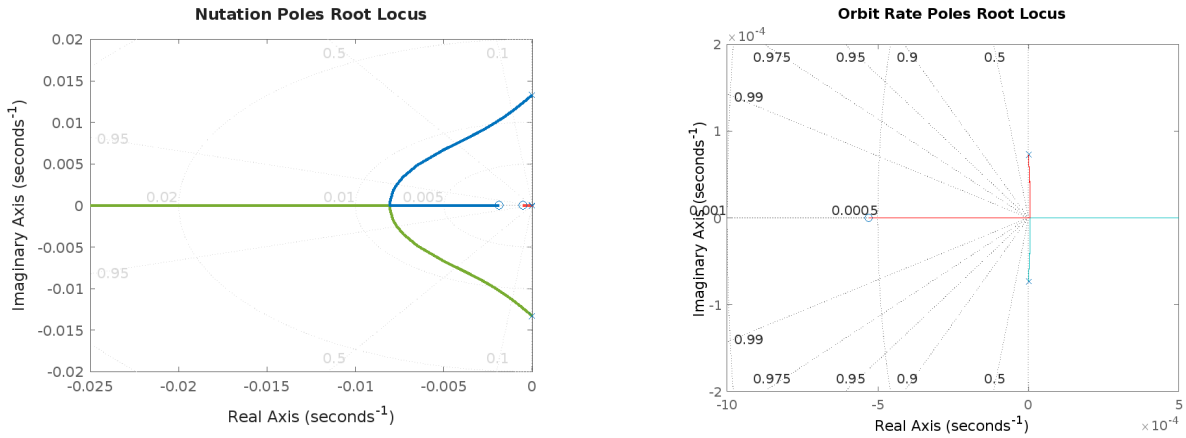


Figure 28: Root Locus of the System with Controller (Nutation and Orbit Poles)

the right half-plane, the orbital rate mode remains undamped. Thus, despite the damping of the nutation frequency response, the undamped orbital rate response prevents BIBO stability.

Addition of a Zero in the Left Half s-Plane

If a zero is added in the left half s-plane, the transfer function of the roll-error actuated controller becomes:

$$G(s) = (\tau s + 1) \tag{84}$$

This introduces a zero at $s = -\frac{1}{\tau}$, where $\tau > 0$.

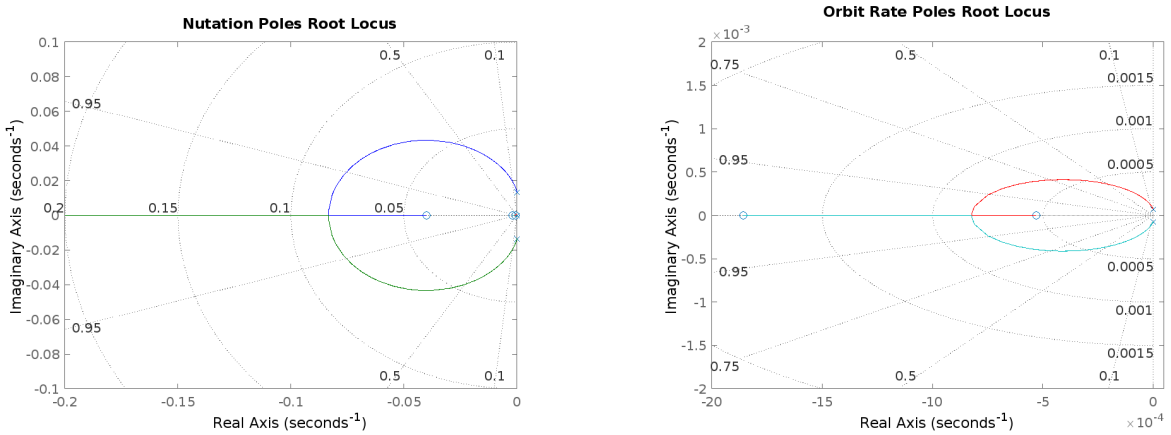


Figure 29: Root Locus of the System with Left Half-Plane Zero (Nutation and Orbit Poles)

Figures 29 show the root locus of the system with an added zero in the left half s-plane. Both subfigures represent the same root locus, zoomed for clarity. The system's orbital modes now move toward the plant's zeros in the left half-plane, damping the orbital frequency response and ensuring BIBO stability for the orbital response.

The added zero also pulls the nutation frequency poles into the left half-plane, damping the nutation frequency response. Thus, with the left half-plane zero, the system becomes BIBO stable for both the orbital and nutation frequency responses.

6.2 Controller Transfer Function and Gains Calculation

The transfer function between the control torque to the roll error can be given as [10]

$$\frac{Control\ Torque}{Roll\ Error} = T_c \times (\tau s + 1) \tag{85}$$

The control torque command equation, to implement the roll-error actuated control is then :

$$T_{cx} = -(k_x \phi_{error} + k_{xd} \dot{\phi}_{error}) \tag{86}$$

$$T_{cz} = -aT_{cx} \tag{87}$$

The equations of dynamics together with the PD control torque commands give :

$$T_{dx} = I_x \ddot{\phi} + \omega_0 . h . \phi + h\dot{\psi} + k_x \phi_{error} + k_{xd} \dot{\phi}_{error} \tag{88}$$

$$T_{dz} = I_z \ddot{\psi} + \omega_0 . h . \psi - h\dot{\phi} + a . k_x \phi_{error} + a . k_{xd} \dot{\phi}_{error} \tag{89}$$

Analyzing the dynamics in the frequency domain using the characteristic equation $\Delta(s)$,

$$\Delta(s) I_x I_z = s^4 I_x I_z + s^3 I_z k_{xd} + s^2 \left[\omega_0 h (I_x + I_z) + I_z k_x + h^2 + ahk_{xd}\right] + s\left[ahk_x + \omega_0 hk_{xd}\right] + \left[(\omega_0 h)^2 + \omega_0 hk_x\right]$$

Stability analysis of the characteristic equation provides the stability conditions as

- h > 0 : the momentum wheel must be oriented towards the negative y-axis.
- a < 0 : the torque along x and z needs to be proportional and opposite
- $k_{xd} > 0$

Steady state analysis using the final value theorem gives the steady state errors as

$$\phi_{ss} = \frac{T_{dx}}{\omega_0 h + k_x} \tag{90}$$

$$\psi_{ss} = \frac{T_{dz}}{\omega_0 h} + \frac{aT_{dx} k_x}{\omega_0 h (k_x + \omega_0 h)} \tag{91}$$

With known and estimated external disturbances, and with limits on the maximum permitted steady state error in yaw, we get the needed minimum momentum bias h. k_x is calculated using the maximum permitted steady state roll error and h. Also, in practical cases, like the offset thruster actuation, a < 1.

To calculate k_{xd}, we compare the determinant of the characteristic equation. Without loss of generality, it is assumed that this determinant consists of two second-order damped poles $\omega_{n1,n2}$ with damping coefficients $\xi_{1,2}$

$$\Delta(s) = (s^2 + 2\xi_1\omega_{n1}s + \omega_{n1}^2)(s^2 + 2\xi_2\omega_{n2}s + \omega_{n2}^2) \tag{92}$$

Simplifying equation (92) and equating the coefficients of the characteristic equation formulations, we get k_{xd}, ω_{n1}, ω_{n2} for the desired value of damping coefficients. The poles ω_{n1}, ω_{n2} correspond to the new closed-loop modal frequencies.

6.3 Selection of Damping Coefficients

6.3.1 Equal Damping Coefficients

Defining the following parameters, for $\xi_1 = \xi_2 = \xi$

$$A = \sqrt{\frac{\omega_0^2 h^2 + \omega_0 h k_x}{I_x I_z}} \tag{93}$$

$$B = \frac{1}{2I_x\xi} \tag{94}$$

$$C = \frac{\omega_0 h(I_x + I_z) + I_z k_x + h^2}{I_x I_z} \tag{95}$$

The system of equations formed by equating the coefficients of the characteristic equation give the control and response parameters as

$$k_{xd} = \sqrt{\frac{(C + A(2 - 4\xi^2))k_x}{k_x B^2 - 2\xi AB + (h\omega_0/(I_x I_z))}} \tag{96}$$

$$a = \frac{2\xi AB k_{xd} I_x I_z - \omega_0 k_{xd} h}{k_x h} \tag{97}$$

$$\omega_{n1} = \frac{k_{xd}B + \sqrt{k_{xd}^2 B^2 - 4A}}{2} \tag{98}$$

$$\omega_{n2} = \frac{A}{\omega_{n1}} \tag{99}$$

6.3.2 Different Damping Coefficients

The nutation frequency is about 3 orders greater than the orbital frequency. Thus, we can achieve the desired time constant of the closed loop even with a lower damping factor for the nutation frequency. This reduces excess thruster firing.

The earth sensor senses the roll and not the roll rate. Roll rate is calculated from the noisy roll measurements. Reducing the damping factor for the nutation frequency also reduces the derivative gain of the controller, which reduces the controller's response to noise, rather than the measurement.

k_x	k_{xd}	ω_{n1}	ξ_1	ω_{n2}	ξ_2	T_{CX}/N_{ES}	T_{CZ}/N_{ES}
4.27e-3	42.8	8.49e-5	0.7	0.07	0.52	2.09	1.86
4.27e-3	4.28	1.32e-4	0.65	0.03	0.08	0.19	0.17
4.27e-3	0.43	1.42e-4	0.65	0.02	4.25e-3	0.01	0.02

Table 2: Reduction of noise amplification with decreasing derivative gain k_{xd} [9]. The parameters k_x, k_{xd}, ω_{n1}, ξ_1, ω_{n2}, and ξ_2 characterize the dynamic behavior of the system under different gain settings. The results show how noise amplification (T_{CX}/N_{ES} and T_{CZ}/N_{ES}) decreases with k_{xd}.

6.4 Techniques of finding the Controller Gains

6.4.1 Iterative Approach

For an initial assumption of the value of a ($a_{guessed}$), and known system capabilities requirement, equations (90) and (91) give

$$k_x = \frac{T_{dx}(\psi_{ss}/\phi_{ss}) - T_{dz}}{\psi_{ss} - a_{guess}\phi_{ss}} \tag{100}$$

$$h = (\frac{T_{dx}}{\phi_{ss}} - k_x)\frac{1}{\omega_0} \tag{101}$$

Equating the coefficients of the characteristic equation formulations, we get k_{xd}, ω_{n1}, ω_{n2} and $a_{calculated}$. Since the initial calculation of h and k_{xd} involves the guessed parameter (a_{guess}), which might not be equal to the value of the parameter calculated ($a_{calculated}$) in the first calculation we interatively calculate the parameters till a_{guess} is within a certain pre-defined allowed tolerance of $a_{calculated}$. The algorithm is illustrated in Figure 30.

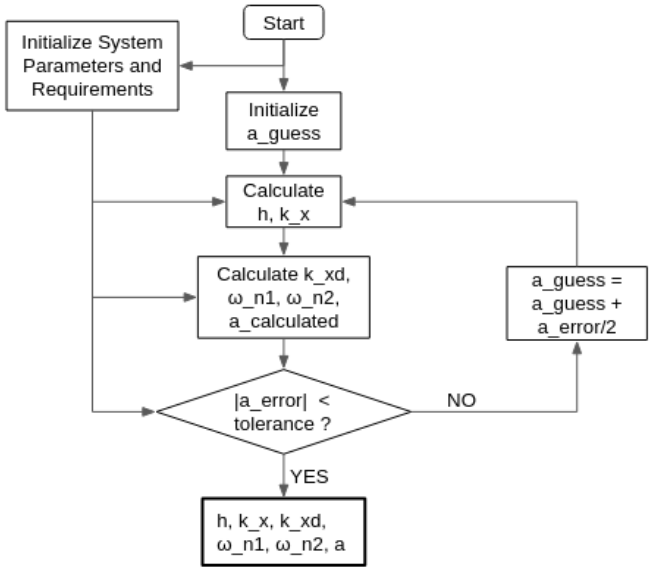


Figure 30: Parameter Calculation Algorithm

Limitations : Non-linear equations can exhibit complex behaviors such as multiple solutions, oscillations, or even chaotic dynamics. The convergence to a solution is also not guaranteed. The choice of initial guess can greatly affect the success of iterative methods. Non-linear equations often have multiple local solutions or regions of attraction. If the initial guess is not sufficiently close to the desired solution or falls within a different region of attraction, the iterative method may converge to a different solution or fail to converge altogether. The convergence rate of iterative methods for non-linear equations can be slow. Depending on the characteristics of the equation and the initial guess, the iterative process may require a large number of iterations to reach a satisfactory solution. This can result in significant computational time and resources, especially for complex and computationally demanding problems. The computational cost of solving highly non-linear equations iteratively can be significant. Each iteration typically involves evaluating the equation and updating the guess, which can be computationally intensive, particularly for large-scale problems.

6.4.2 Fixing value of h

A better approach of solving is to fix a value of h, and find k_x using the steady state condition on roll. We then solve the 4 equations to get k_{xd}, ω_{n1}, ω_{n2}, and a. We calculate the steady state error in yaw and see if it satisfies the required steady state condition on yaw. For a fixed h,

$$k_x = \frac{T_{dx}}{\phi_{ss}} - \omega_0 h \tag{102}$$

We then solve the 4 equations to find k_{xd}, ω_{n1}, ω_{n2}, and a_ψ. We use these parameters to calculate the steady state error in yaw $\psi_{ss,c}$.

$$\psi_{ss,c} = \frac{T_{dz}}{\omega_0 h} + \frac{T_{dx} a_\psi k_x}{\omega_0 h(\omega_0 h + k_x)} \tag{103}$$

For the desired steady state conditions [11] :

$$\phi_{ss,d} < 0.05^\circ; \qquad \psi_{ss,d} < 0.4^\circ \tag{104}$$

spacecraft moments of inertia defined as :

$$I_x = 1475.8425\,\text{kg m}^2; \qquad I_z = 1532.96\,\text{kg m}^2 \tag{105}$$

maximum disturbance torques :

$$T_{dz} = T_{dx} = 5 \times 10^{-6}\,\text{Nm} \tag{106}$$

and orbit rate defined for a geostationary orbit :

$$\omega_0 = \frac{2\pi}{86400}\,\text{rad/s} = 7.2722 \times 10^{-5}\,\text{rad/s} \tag{107}$$

Input Torque	k_x	k_{xd}	ω_{n1}	ξ_1	ω_{n2}	ξ_2	a_ψ	$\phi_{ss,c}$	$\psi_{ss,c}$
15 Nms	4.64e-3	21.746	1.14e-4	0.7	1.46e-2	0.5	0.768	0.05°	0.426°
	4.64e-3	3.53	1.515e-4	0.65	1.1e-2	0.1	0.717	0.05°	0.415°
	4.64e-3	1.867	1.566e-4	0.65	1.06e-2	0.05	0.717	0.05°	0.415°
	4.64e-3	0.614	1.61e-4	0.65	1.034e-2	0.01	0.717	0.05°	0.415°
20 Nms	4.275e-3	31.577	9.023e-5	0.7	2.13e-2	0.5	0.980	0.05°	0.341°
	4.275e-3	4.604	1.3e-4	0.65	1.475e-2	0.1	0.897	0.05°	0.329°
	4.275e-3	2.3472	1.357e-4	0.65	1.414e-2	0.05	0.894	0.05°	0.328°
	4.275e-3	0.673	1.404e-4	0.65	1.367e-2	0.01	0.891	0.05°	0.328°
25 Nms	3.912e-3	44.2778	7.176e-5	0.7	2.99e-2	0.5	1.259	0.05°	0.293°
	3.912e-3	5.7483	1.15e-4	0.65	1.873e-2	0.1	1.103	0.05°	0.276°
	3.912e-3	2.8533	1.21e-4	0.65	1.776e-2	0.05	1.094	0.05°	0.275°
	3.912e-3	0.7445	1.26e-4	0.65	1.704e-2	0.05	1.086	0.05°	0.274°
30 Nms	3.548e-3	62.815	5.533e-5	0.7	4.25e-2	0.5	1.687	0.05°	0.268°
	3.548e-3	6.9743	1.024e-4	0.65	2.3e-2	0.1	1.350	0.05°	0.241°
	3.548e-3	3.3832	1.093e-4	0.65	2.15e-2	0.05	1.328	0.05°	0.239°
	3.548e-3	0.82375	1.151e-4	0.65	2.043e-2	0.01	1.310	0.05°	0.238°

Table 3: Parameter settings and results for different input torque values, showing how the parameters affect the system's response.

6.5 Offset Angle Constraint

Having discussed the conditions on the poles of the controlled system, we analyze its zeroes. The zero added by the controller has already been discussed above. The zeroes of the system can be found out by analyzing

$$H_{roll}(s) = \cos\alpha \, \frac{I_z s^2 + h \, \tan\alpha \, s + \omega_0 h}{I_x I_z (s^2 + \frac{h^2}{I_x I_z})(s^2 + \omega_0^2)} \tag{108}$$

The zeroes $z_{1,2}$ are given by

$$z_{1,2} = \frac{-h \, \tan\alpha \, \pm \sqrt{h^2 \, \tan^2\alpha \; - \; 4\,\omega_0\,h\,I_z}}{2 \, I_z} \tag{109}$$

The modes of the system corresponding to the orbit frequency poles at $\pm j\omega_0$ are damped by attraction to two zeros in the left half s-plane whose location depends mainly on α. To achieve good damping of the orbital dynamic mode it is desirable that these zeros be located on, or very near to, the negative real axis of the s-plane.

Since ω_0, h, I_z are positive,

$$|h \, tan\alpha| > |\sqrt{h^2 \, \tan^2\alpha \; - \; 4\,\omega_0\,h\,I_z}\,| \tag{110}$$

The above mentioned requirement of the zeroes of the plant being negative and real (which enable overdamping the orbital rate poles) can be achieved by just ensuring that the zeroes of the plant are real. This can be done by appropritate selection of α. Dougherty [10] discusses that to prevent unnecessary expenditure of gas due to "ringing" these zeros are chosen to be negative by setting the offset angle α so that

$$\alpha \geq \tan^{-1} \; 2\sqrt{\frac{\omega_0 I_z}{h}} \tag{111}$$

Although the above calculations provide a constraint on the offset angle to enable overdamping of the system response, it is often not used since the system is never overdamped. As seen from the controller parameter calculation table **??**, the damping ratio is always less than 1. Furthermore, the offset angle for the calculated parameters is not near the lower limit of the offset angle calculated in this section, in agreement with Dougherty [10].

6.6 Discrete-Time Simulation Implementation

The z-transform is used to deal with discrete-time implementation

$$X[z] = Z\{x[k]\} = \sum_{k=0}^{\infty} x[k] z^{-k} \tag{112}$$

in which we use the Time-Shifting Property

$$Z\{x[k-n]\} = z^{-n} X[z] \tag{113}$$

where $Z\{x[k]\} = X[z]$.

The implementation PD Control is done using bilinear transform.

$$u(s) = K_d s + K_p \tag{114}$$

in the discrete-time domain is given by the transformation

$$s \rightarrow \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}} \tag{115}$$

where T_s is the sampling time. This gives us the PD control in its discrete form

$$\frac{U[z]}{E[z]} = \frac{(K_p + K_d(2/T_s) + (K_p - K_d(2/T_s))z^{-1}}{1 + z^{-1}} \tag{116}$$

Implemention of the discrete-time PD Control is then done as

$$\boxed{u[k] = -u[k-1] + (K_p + K_d(2/T_s))\, e[k] + (K_p - K_d(2/T_s))\, e[k-1]} \tag{117}$$

7 Sensor Modeling and Noise Filters

Mathematical modeling of sensor noise ensures realistic simulations by emulating sensor behavior and characteristics. This approach enhances simulation fidelity, enables control system performance evaluation, and analyzes the impact of sensor data on overall system behavior. Sensor noise, arising from environmental factors, electronic interference, or sensor limitations, distorts measurements and reduces accuracy. To address this, low-pass filters are commonly used, allowing low-frequency signals while attenuating high-frequency noise. This improves signal quality and enhances the accuracy of simulated sensor measurements.

7.1 Sensor Model

Earth sensors determine satellite roll and pitch by sensing the Earth's horizon relative to the satellite body frame. The sensor noise model, based on specifications from Markland [12] and Iwen [11], includes:

- Quantization : 0.02 °
- Statistical Noise : 0.012 ° RMS

Measurements are accurate only up to the sensor's quantization, with higher-order details lost as values are rounded to the nearest multiple of the quantization. The simulation algorithm 1 processes true input values to generate outputs influenced by quantization and noise, replicating real sensor behavior.

Algorithm 1: Quantized Sensor Measurement Model

Input: Signal amplitude x_{in}, quantization step Δq

Output: Quantized measurement x_{meas}

$$\text{sgn}(x_{in}) \leftarrow \begin{cases} -1, & x_{in} < 0 \\ +1, & x_{in} \geq 0 \end{cases}$$

$|x_{in}| \leftarrow \text{magnitude}(x_{in})$

$n_q \leftarrow \left\lfloor \frac{|x_{in}|}{\Delta q} \right\rfloor$

$r_q \leftarrow |x_{in}| \bmod \Delta q$

if $r_q \geq 0.5\Delta q$ **then**

  $n_q \leftarrow n_q + 1$

$x_{det} \leftarrow \text{sgn}(x_{in}) \cdot n_q \cdot \Delta q$

$x_{meas} \leftarrow x_{det} + \eta$, where $\eta \sim \mathcal{N}(0, \sigma^2)$

return x_{meas}

Sensor noise is modeled as a zero-mean, uncorrelated random sequence with specific variance, effectively representing high-frequency noise [11]. Typically, it is represented as Gaussian White Noise, characterized by independent and identically distributed (IID) random variables. The RMS value defines the standard deviation of the Gaussian distribution. Figure 31 illustrates an instance of Gaussian White Noise with a standard deviation of 0.012. The nature of noise as white can be inferred from its autocorrelation plot. For white noise, characterized by random, uncorrelated

fluctuations, the autocorrelation values remain close to zero for all lags. Figure 32 presents the autocorrelation plot of the generated noise, confirming its white noise characteristics.

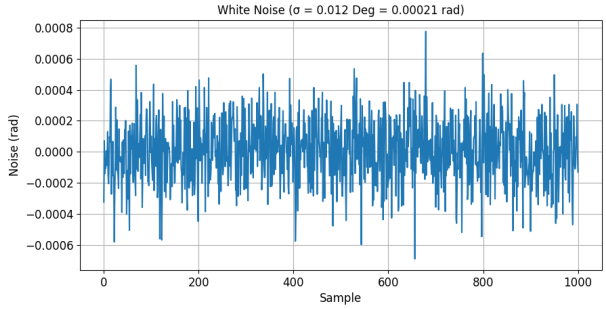


Figure 31: Gaussian White Noise

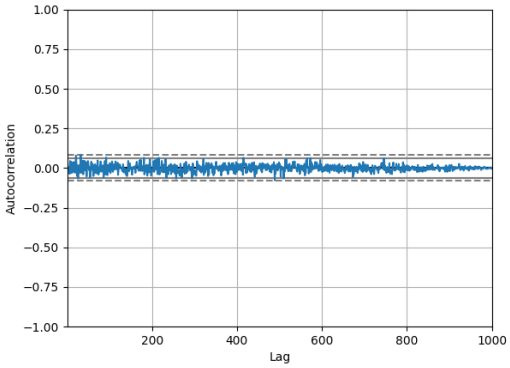


Figure 32: Autocorrelation Plot

7.2 Output Rate

Earth sensors often provide output in sample-and-hold form with typical sampling intervals ranging from T $\approx$ 0.25 s to T = 1.0 s. For a simulation sampling rate of T = 0.5 s, the output frequency is:

$$f_{output} = \frac{1}{T} = 2\,Hz \approx 12.566\,rad/s \tag{118}$$

This frequency also corresponds to the noise addition rate. Low-pass filters are used to smooth measurements and mitigate noise, with the filter selection based on the system's bandwidth without noise.

7.3 System Bandwidth

The system's time response is characterized by the closed-loop response frequencies. For the final simulation parameters, we have:

$$\omega_1 = 1.357 \times 10^{-4}\,\text{rad/sec} \tag{119}$$

$$\omega_2 = 1.414 \times 10^{-2}\,\text{rad/sec} \tag{120}$$

A common guideline is to set the filter corner frequency $\omega_c = 10\omega_2$, ensuring $\omega_c \geq 7.5\omega_2$ to avoid excessive phase shift [11]. Therefore, we choose:

$$\omega_c \approx 1.414 \times 10^{-1}\,\text{rad/sec} \tag{121}$$

This cut-off frequency is used to filter sensor noise. Given that our frequencies of interest are lower than the noise frequency, a low-pass filter with bandwidth BW = ω_c is applied to remove noise from the roll measurement.

7.4 Filter Selection

Recursive filters (IIR type) are efficient and fast but offer less performance and flexibility compared to Windowed filters (FIR type). While IIR filters are faster by an order of magnitude, they are often preferred in real-time systems. The Butterworth low-pass filter is ideal for achieving smooth, linear phase response, balancing passband ripple and stopband attenuation [13] [14]. The cut-off frequency for our simulation is calculated as mentioned earlier, and a first-order filter is sufficient for the desired noise attenuation [11].

7.5 Filter Implementation

The transfer function for an N-order recursive low-pass filter is:

$$H(z) = \frac{B(z)}{A(z)} = \frac{b'_0 + b'_1 z^{-1} + \cdots + b'_N z^{-N}}{a'_0 + a'_1 z^{-1} + \cdots + a'_N z^{-N}} \tag{122}$$

By normalizing the coefficients with respect to a_1, the transfer function becomes:

$$H(z) = \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_N z^{-N}}{1 + a_1 z^{-1} + \cdots + a_N z^{-N}} \tag{123}$$

This corresponds to the difference equation:

$$y[n] = b_0 x[n] + b_1 x[n-1] + \cdots + b_N x[n-N] - a_1 y[n-1] - \cdots - a_N y[n-N] \tag{124}$$

where $x[n]$ is the input and $y[n]$ is the output sequence.

The coefficients $a_1 \ldots a_n$ and $b_0 \ldots b_n$ are computed using the 'butter()' function from the 'scipy.signal' module, with the normalized cut-off frequency, filter order, sampling frequency, and filter type as inputs. Figures 33 and 34 illustrates the effect of a first-order filter, which achieves the desired noise attenuation [11]. Higher-order filters can be analyzed to assess the trade-offs in filter order selection.

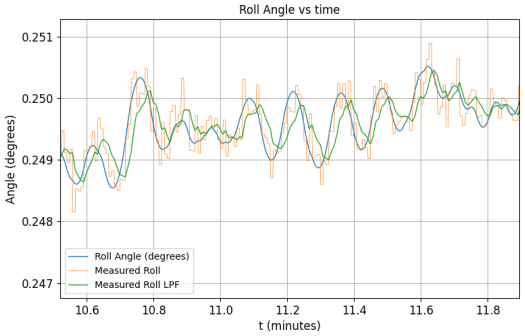


Figure 33: First Order Butterworth LPF

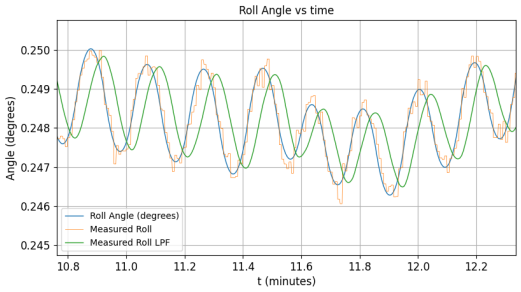


Figure 34: Second Order Butterworth LPF

8 Reaction Thruster Actuation

Reaction jet control systems (RCS) are a significant advancement in spacecraft engineering, enabling precise attitude control and orbit adjustments. These systems convert control commands into thruster activation commands, but the algorithm faces complications due to two factors [9]:

1. Reaction thrusters are nonlinear; the torque produced depends on the activation duration.
2. Thrusters produce only one-signed torques, requiring opposite thrusters for reverse torque about the same axis.

These factors complicate the transformation from body torque commands to thruster activation.

Pulse-Width Pulse-Frequency Modulation (PWPFM) converts control signals to thruster commands through combined width and frequency modulation (Fig. 35). The system comprises a Schmitt trigger with deadzone and hysteresis, a first-order lag network, and negative feedback. Advanced thruster models incorporate start time T_{s}, shutdown time T_{sd}, and centroid time difference T_{cd} between command and thrust (Fig. 36).

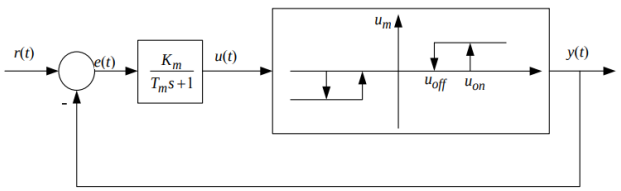


Figure 35: Pulse-Width Pulse-Frequency Modulator [15]

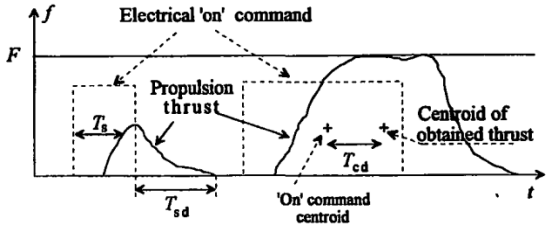


Figure 36: Time Domain Behaviour of Thruster Pulse [9]

8.1 Reaction Thruster Set-up

With the offset thruster setup shown in Figure 26, the control torques are given as:

$$M_x = M\cos\alpha, \tag{125}$$

$$M_z = -M\sin\alpha = (-\tan\alpha)M_x. \tag{126}$$

The offset thruster arrangement eliminates the need for separate calculations of control torques about body axes. By accounting for the thruster offset, the required thruster activations can be directly determined. The PWPF modulator is highly sensitive to its parameters, requiring careful selection to meet system requirements (such as Minimum Pulse Width, Roll Deadband, Fuel Consumption). Furthermore, the parameters lack analytical determination methods due to their complex nonlinear interactions. Current implementations rely on numerical simulations and empirical tuning, without a rigorous theoretical framework for optimal parameter selection.Thus the parameters were chosen based on an extensive review of literature [16] [17] [18] [15] [19] [20] [21] [22].

9 Simulation Results

9.1 Reproducing Results by Dougherty [10]

The simulink model, shown in Figures 37 to 39, simulates the coupled roll-yaw equations.

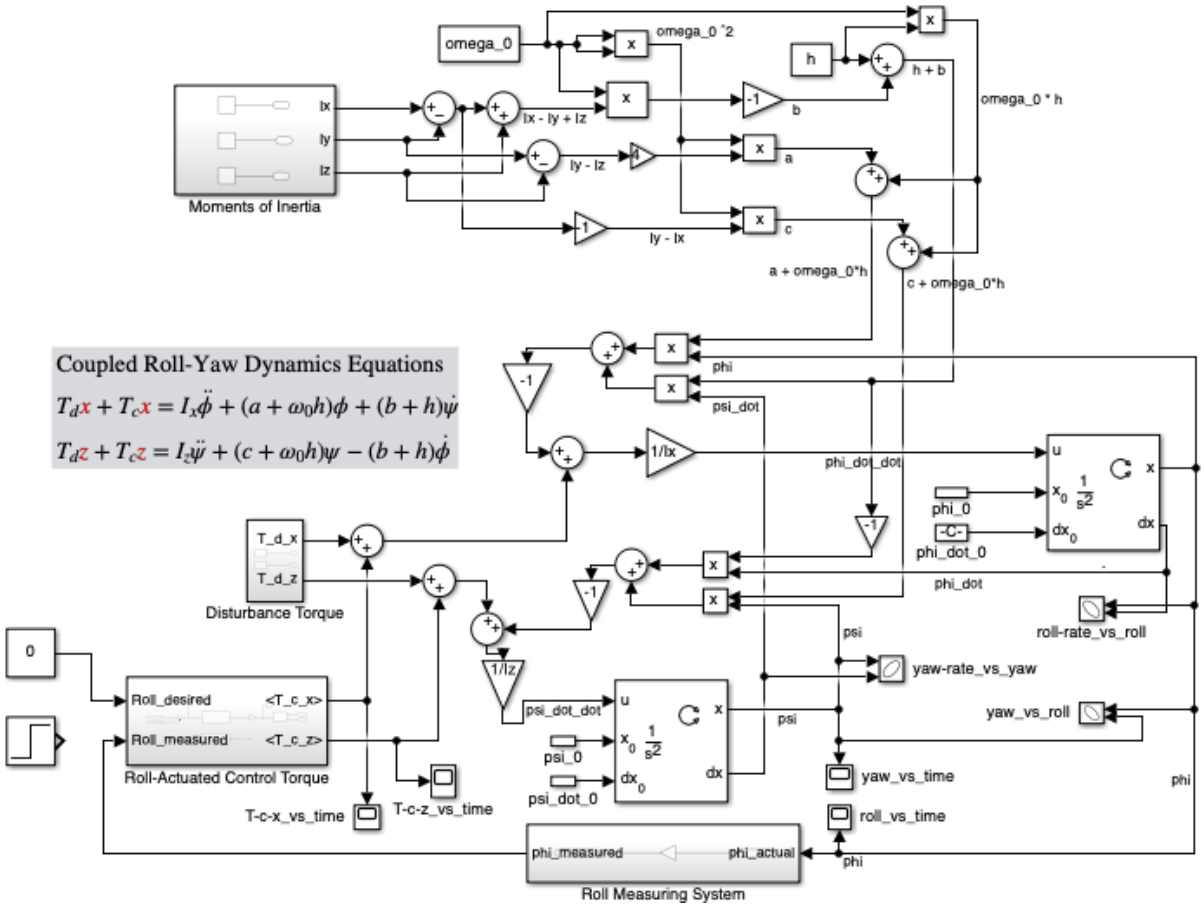


Figure 37: Simulink System Model

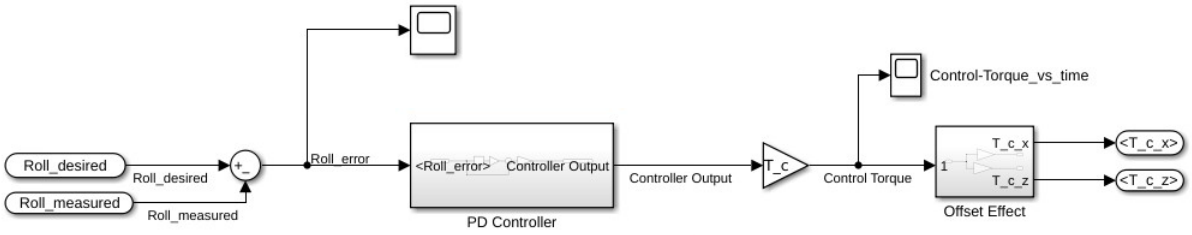


Figure 38: Roll-Actuated Control Torque

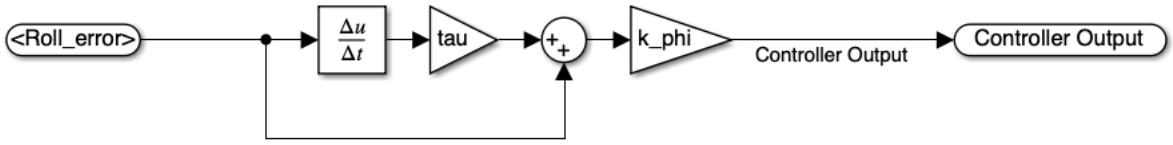


Figure 39: PD Controller

It assumes exact roll information to compute the control output, with gains specified by Dougherty [10]. The control output is idealized, differing from real-world conditions.

System Parameters

Table 4 summarizes the key parameters used in the simulation, including physical constants, control settings, and initial conditions.

Parameter	Description	Value
$\boldsymbol{I_x, I_z}$	Principal moments of inertia	1355.818 kg·m^2
$\boldsymbol{\tau}$	Controller time constant	25 s
$\boldsymbol{K_\phi}$	Control gain	62.3676 kg·m/rad
$\boldsymbol{\omega_0}$	Orbital frequency	0.73×10^{-4} rad/s
$\boldsymbol{h}$	Momentum wheel angular momentum	203.3727 N·m·s
$\boldsymbol{\phi_0}$	Initial roll angle	0.0174533 rad
$\boldsymbol{\psi_0, \dot{\phi}_0, \dot{\psi}_0}$	Initial yaw, roll rate, yaw rate	0 rad, 0 rad/s

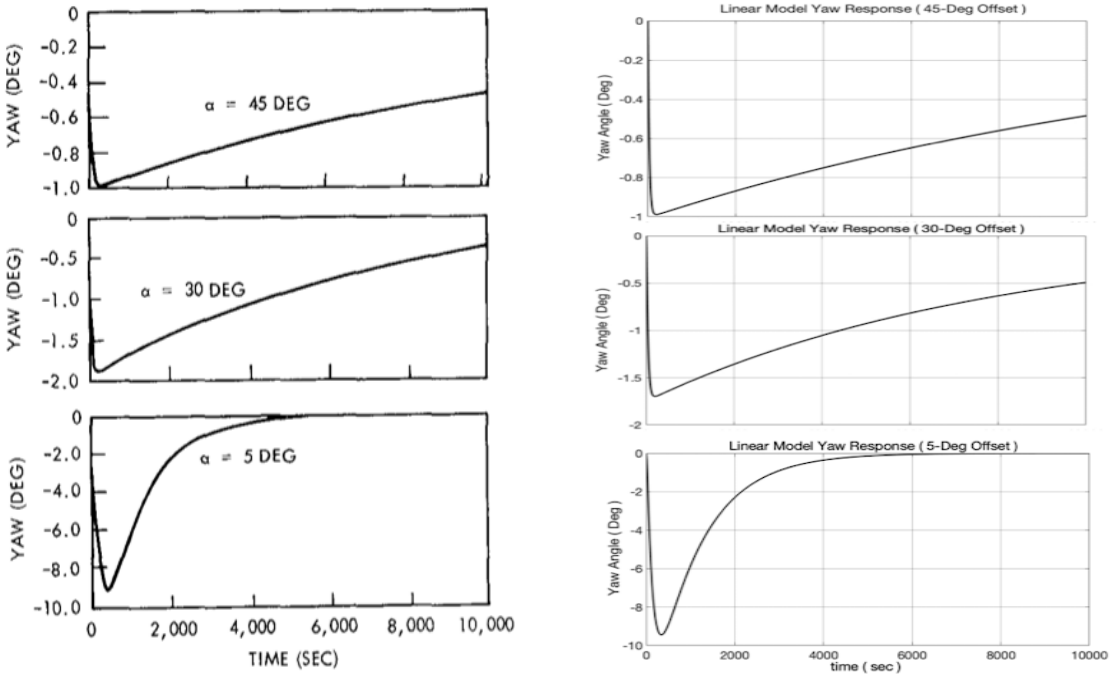
Table 4: Simulation Parameters

Model Limitations

- Sensor noise and quantization effects neglected
- Controller parameter selection criteria unspecified
- External disturbances not modeled
- Ideal actuator response assumed
- Sensor sampling effects not considered

Model Verification

Figure 40 compares the simulation results with Dougherty's original work [10], demonstrating model validity.



(a) Results from Dougherty [10]

(b) Results obtained from Simulink model

Figure 40: Comparison of Results for Model Verification

9.2 Final Simulation

The simulation limitations of the Simulink model were addressed in the final system simulation. It incorporates earth sensor characteristics, including noise and quantization, with noise filtered by a low-pass filter. The earth sensor operates at a fixed output rate in sample-and-hold mode. Control parameters are selected, and the control output is converted to reaction thruster commands using pulse-width pulse-frequency modulation.

- the roll sensor has a quantization of 0.02 deg, therefore the deadband for actuation must be greater than that, otherwise the system is responding to sensor noise.
- the minimum on time for thruster is 0.05 seconds, thus the simulation time must atleast be that much to account for it
- the offset thruster causes variation in yaw!!

Parameter	Value	Description
Iz	1532.96 kg-m^2	Yaw Moment of Inertia of the s/c
Ix	1475.8425 kg-m^2	Roll Moment of Inertia of the s/c
Iy	886.2425 kg-m^2	Pitch Moment of Inertia of the s/c

Table 5: System Parameters

Parameter	Value	Description
T_{output}	0.5 sec	Earth Sensor Output Period
σ_{sensor}	0.012 Deg	Earth Sensor Noise
q_{sensor}	0.02 Deg	Earth Sensor Quantization

Table 6: Roll Sensor Parameters

Parameter	Value	Description
ϕ_{max}	0.05 Deg	Maximum Steady State Roll Error
ψ_{max}	0.35 Deg	Maximum Steady State Yaw Error
$T_{d_{max}}$	5e-6 N-m	Maximum Disturbance Torque (both T_{d_x}, T_{d_z})

Table 7: System Capabilities Requirement

Parameter	Value	Description
ξ_{orbit}	0.65	Orbit-rate Damping Coefficient
$\xi_{nutation}$	0.05	Nutation Damping Coefficient
h	20 Nms	Momentum Bias
k_x	4.275e-3 Nm	Proportional Feedback Gain
k_{xd}	2.3472 Nms	Derivative Feedback Gain
α	41.8 Deg	Thruster Offset Angle
T_c	100	Thruster Gain Parameter

Table 8: Control System Parameters

Parameter	Value	Description
K_m, τ_m	2, 0.2	PWPFM Filter Parameters
U_{on}, U_{off}	0.05, 0.02	PWPFM Schmitt Trigger Parameters
U_m	0.01 N	Thruster Output Force
d	1 m	Effective Moment Arm of Offset Thrusters
t_{min}	0.05 sec	Minimum Thruster On-Time
T_{min}	0.005 Nms	Minimum Impulse Bit (t_{min} * U_m * d)

Table 9: Actuation Parameters

Results

10 Conclusion

Target Pointing Guidance Commands

Latitudinal perturbations in the coordinates of the sub-satellite point (SSP) are significantly larger than longitudinal ones. Orbital inclination causes both latitudinal and longitudinal perturbations, creating a figure-eight ground trace. Eccentricity introduces only longitudinal perturbations, while deviations in the semi-major axis shift the longitude of the ascending node. The 3-2-1 rotation sequence simplifies the generation of pointing commands, making it an effective choice for attitude definition. Pitch pointing commands are two orders of magnitude smaller than roll and yaw commands. Consequently, pitch control, implemented through changes in the momentum wheel, has negligible impact on overall system dynamics.

Bias Momentum Satellite Dynamics and Control

The orientation of the momentum wheel is critical for enabling roll-yaw coupling and ensuring effective control. Pitch dynamics are decoupled and can be independently managed using momentum wheel adjustments or additional low-thrust thrusters. However, the momentum wheel increases system complexity by introducing a second response frequency, resulting in both precession and nutation frequencies. Approximations of nutation frequencies remain valid for $h > 1$ Nms. A Proportional-Derivative Controller is effective in damping system modes associated with these frequencies.

Offset Thrusters Control Implementation

Bias momentum enables the use of offset thrusters for simultaneous torque application about two body axes, simplifying control. Precise tuning of Pulse-Width Pulse-Frequency Modulation (PWPFM) parameters is essential to achieve accurate and reliable thruster control.

10.1 Direction for Future Work

- **Advanced Control Techniques:** Explore adaptive control, model predictive control, or machine learning-based algorithms to further optimize pitch and yaw control, improving robustness in the presence of uncertainties and disturbances.
- **Real-Time Testing:** Conduct extensive real-time testing in simulated environments that closely replicate the operational conditions the spacecraft will encounter, such as gravitational forces, atmospheric drag, and sensor noise, to refine control algorithms.
- **Disturbance Rejection:** Investigate methods to improve disturbance rejection capabilities, particularly for external forces like solar radiation pressure, magnetic fields, or unexpected impacts that might affect spacecraft orientation.
- **Fault Tolerance:** Develop fault detection and recovery strategies for the control system, ensuring continued performance under hardware failures or other unexpected issues.
- **Optimization of Control Parameters:** Use more sophisticated optimization techniques (e.g., genetic algorithms, particle swarm optimization) to fine-tune control parameters, improving efficiency and reducing fuel consumption while ensuring stability.

- **Integration with Guidance and Navigation Systems:** Ensure that the pitch and yaw control system is seamlessly integrated with the spacecraft's overall guidance, navigation, and control (GNC) systems for coordinated operations, ensuring mission objectives are met efficiently.
- **Scalability and Modular Design:** Design control systems with scalability in mind to adapt to spacecraft with varying sizes, mission profiles, and power constraints, ensuring broad applicability across different projects.
- **Testing with Real Hardware:** If feasible, transition from simulation to hardware-in-the-loop (HIL) testing to validate the control algorithms on actual spacecraft components, ensuring real-world applicability and reliability.

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